Modeling

Table of Contents

What the model decided

Six things. Each is a decision the project took because of the number beside it, not a description of a simulation. Everything after this page is the working.

1 · Is the AND gate really a *product* of its two inputs? Yes — at the endpoint. The two-way interaction is zero within error at 24 h, and the gate predicts any held-out corner to within 2.2 %. At 6 and 9 h the same prediction is wrong by up to 4.4×. → *Licensed the entire safety chapter to factorise the gate, and forbade extrapolating mid-course ON/OFF ratios* (§ 1.1, § 4.1).

2 · Where, other than a tumour, does the circuit switch on? The colonic mucosa, at 21.85× — and only because the *native* lldPRD repressions hold the gut lumen off. On the designed arms alone, all three luminal sites sit **below 1.0**. → *Found a bottleneck nobody had looked at: last cycle's site list had no lumen in it at all* (§ 2.1).

3 · Can we "improve" the lactate sensor by removing its glucose/anoxia repression? No — that would destroy the safety argument. Luminal discrimination falls from 87–228× to 0.87–2.28×. → 🔴 *A design decision: do not swap in a de-repressed variant* (§ 2.3).

4 · Below what lesion size can the hypoxia arm not fire, for *any* parameters? ≈178 µm. Pure geometry, zero fitted parameters — and it lands inside the 160–200 µm bracket Thomlinson & Gray measured histologically in 1955. → *The hardest safety statement in the project, and the only parameter an independent measurement confirms* (§ 2.2).

5 · Keep the SsrA degron or drop it? Keep. All three doses clear the decision threshold, and ClpXP saturation means the degron *amplifies* selectivity rather than merely damping output. → *Settled a build question the raw biomass numbers alone would have answered backwards* (§ 2.4).

6 · What dose does a therapeutic effect need? The clinical indication is met below the lowest dose we tested; CT26 needs 1.3–1.8× the highest. But reaching a 150 µm micro-lesion costs 4.95× and leaves only 4.41× of the mucosa margin. → *Set the dose target, and showed micrometastasis is **not** a dose problem — which is why § 3.6 exists* (§ 3.4, § 3.5).

Every number on this page is computed, not transcribed. The figures, the tables and the worked substitutions are generated from the code in this repository, and a test suite of 329 assertions pins each of them to the function that produces it — so the page cannot hold a stale number, and a reader who wants to check one can run it. § 4 then puts the central claims through out-of-sample tests: held-out gate corners, held-out doses, and a 24 h capacity forecast built from truncated curves. All pass.

Two of these results changed our own minds, which is the part we would most like a reader to weigh. § 2.5 records two inherited sensor parameters that turned out to have published measurements disagreeing with them; we adopted the measurements and recomputed every number here. § 5.1 withdraws a selectivity index we had previously published, caught by our own consistency checking rather than by anyone objecting.

What we set out to answer

Two questions decide whether this design is worth building:

> **Safety — where, other than a tumour, does this circuit switch on?**
>
> **Efficacy — what dose would a therapeutic effect require?**

Everything below is organised under those two headings. We do not organise by length scale, because a micro/meso/macro split describes the *model* rather than the *decision*, and every section here exists to settle a decision.

The circuit:

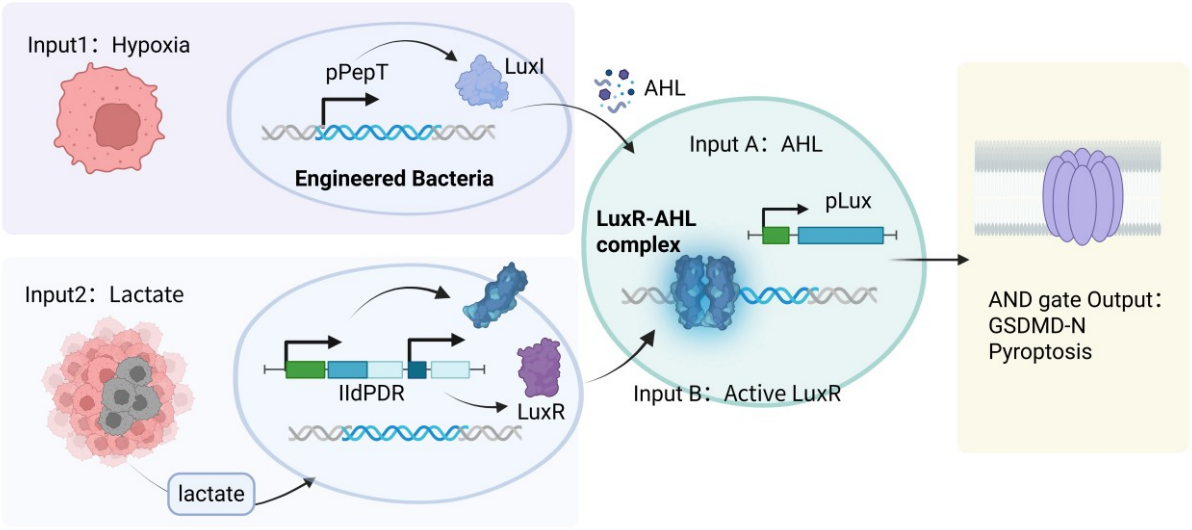


Figure 1. The two-input AND gate. Hypoxia drives LuxI from pPepT; lactate drives LuxR from the native lldPRD operon promoter; only the LuxR–AHL complex opens pLux on the effector. Chassis is *E. coli* Nissle 1917, delivery is oral, and the

indication is pMMR/MSS colorectal cancer. That combination is not ours: orally dosed EcN selectively colonises adenomas in *Apc*$^{Min/+}$ mice and was enriched in tumour over normal tissue in a double-blind trial of patients dosed before resection [1], and EcN replicates at the live/necrotic border of tumours while not colonising spleen or liver at all [2]. The lactate sensor is **BBa_M36021**, the native *E. coli* lldPRD operon promoter [3], a choice that turns out to carry the whole luminal safety argument (§ 2.3). Two things about that part are worth stating up front: it is a 332 bp basic part derived from BBa_K822000, and the Registry currently marks it **deprecated**, which is a documentation status rather than a defect but is the kind of thing a reader should hear from us rather than discover.

How to read this page

Each section states the specific question it answers, then its governing equations with every symbol defined, then the parameters with their units and provenance, then the numbers substituted in, and only then the result. Where a limitation changes a conclusion we say so next to that conclusion, and § 5 collects the rest.

- **Equation numbers** run (1)…(N) in reading order. The source of this page tags each equation with its number in EQUATIONS.md instead, and the build emits the mapping to outputs/equation_index.md, so every formula here can still be traced to the function that evaluates it.
- **Evidence grades**: **calibration** = fitted to our data · **literature prior** = measured by someone else on a comparable system · **scenario** = a value we declared · **derived** = computed from other entries · **protocol** = a threshold fixed before it was used. The literature-prior grade separates "somebody measured this" from "we picked this", which the first four grades ran together.
- **Full parameter census** is in § 6, generated from config/parameters.csv; src/parameters.py resolves each row against the constant the code actually uses and fails the test suite if the two disagree, so this page cannot hold a stale copy.

Notation

The key mathematical notations used in this page are listed in Table 1. Every symbol is also defined where it first appears, in the where list under each section's equations.

Table 1. Notation.

Symbol	Meaning
$f_L(L)$, $f_O(C)$	lactate-arm and hypoxia-arm activity
L, C	lactate concentration (mM), oxygen (%

Symbol	Meaning
	O_2, gas phase)
n_Q	quorum Hill coefficient — the dominant unknown of the whole axis
N	bacterial density (measured carrying capacity)
R_{margin}	gate output at a site relative to the tumour core (the “margin”)
R	lesion radius (μm)
λ	diffusion screening length (μm), $\lambda = \sqrt{D/\delta}$
Ψ_{centre}	normalised AHL at a lesion centre
d	dose (co-culture OD)
b, A	exponent and amplitude of the net dose–kill power law
δ	effector degradation rate (h^{-1})
Ψ	immune pressure number

1. Calibration

1.0 What our wet lab measured, and what each measurement decides

Everything quantitative on this page is anchored to **391 measurements our own team ran this cycle**, in four blocks. Nothing in this chapter is inherited and nothing in it is a literature value; the whole chapter is the wet lab’s data and what it forces the model to say.

Table 2. The experimental blocks, and the decision each one settles.

Block	What was measured	Measurements	What it decides
1–2	OD_{600} growth curves: 2 constructs (GSDMD, GSDMD–SsrA) × 4 conditions (21/1 %	**192**	the carrying capacity K at every site (§ 2.1) and the chassis cost of the SsrA tag (§ 2.4) — fitted

Block	What was measured	Measurements	What it decides
	O_2 × 0/20 mM lactate) × 8 timepoints × 3 replicates		in §1.3
3	reporter fluorescence intensity: 4 corners × 4 timepoints × 3 replicates	**48**	the magnitude of the hypoxia contribution, 1.33× (§1.2)
4	reporter percent-positive: 4 corners × 4 timepoints × 3 replicates	**48**	**the AND-gate interaction test itself** — the one result the entire safety chapter is factored on (§1.1), and the held-out prediction that validates it (§4.1)
5	co-culture killing, dose series: 7 constructs × 3 doses (6/8/10 OD) × 3 replicates, CT26	**62**	the dose–kill power law $b = 2.651$, and through it **every dose number in §3.4 and the whole safety-dose collision in §3.5**
6	co-culture killing, cell-line pairing: 7 constructs × 2 lines (CT26, L929) at 10 OD	**41**	the construct's own selectivity index, 3.3× (§1.4)

Read the last column: **block 4 is what lets the gate be written as a product at all, and block 5 is what turns the model's safety margin into a dose in OD.** Neither is replaceable by a literature value, because no published study used this construct.

Two measurements do more work than their cost suggests, and both are worth saying out loud:

- Running **both constructs in the same growth block** is what makes the first-hour burst interpretable (§ 1.3). The tagged strain is the control for the medium, so "lactate slows growth" is ruled out by the data rather than argued away.
- Running **block 5's seven constructs at the same three doses** is what lets the killing be taken *net of* the gfp carrier (§ 3.4), instead of crediting the effector with the chassis's own effect.

Where the data does not reach, we say so rather than filling the gap: § 5.1 lists what this workbook does not contain, and § 7 costs the experiments that would close each gap.

1.1 Is the AND gate multiplicative?

> **Question.** Read as raw fluorescence the two inputs together give 227× where a multiplicative gate predicts 2768×. Is that 12× shortfall real sub-multiplicativity, or an artefact of a saturated readout? And once the artefact is removed, what does the data actually support?

This matters because every safety result below writes the gate as a product of three arms (Eq. (11)). This section is the measured basis for that factorisation.

Corners: A = 21 % O_2 · B = 1 % O_2 · C = 21 % + 20 mM lactate · D = 1 % + 20 mM.

Model

If per-cell log-expression is normal with condition-dependent mean μ and a common spread σ, and the cytometry gate is a fixed threshold θ, the positive fraction is the standard probit model [4]:

$$p = \Phi\left(\frac{\mu-\theta}{\sigma}\right) \Rightarrow z \equiv \Phi^{-1}(p) = \frac{\mu-\theta}{\sigma} \qquad (1)$$

A fold change between two corners is then

$$\text{fold} = \exp(\sigma\,\Delta z) \qquad (2)$$

θ cancels from any difference; σ **does not**, and Eq. (2) depends on it exponentially.

A multiplicative gate means the two inputs are *additive* on μ, hence additive in probit units, hence the two-way interaction vanishes:

$$I = z_D - z_C - z_B + z_A \overset{!}{=} 0 \qquad (3)$$

$$SE(I) = \sqrt{\sum_{c\in\{A,B,C,D\}} s_c^2/n_c} \qquad (4)$$

where

- μ is the condition-dependent mean of per-cell log-expression.
- σ is the common spread of per-cell log-expression.
- θ is the fixed cytometry threshold for gating positive cells.
- p is the positive fraction of cells above the cytometry gate.
- $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution, and $\Phi^{-1}(\cdot)$ its inverse (the probit function).
- z is the probit-transformed positive fraction.
- fold is the fold change between two experimental corners.
- I is the two-way interaction term, zero for a multiplicative gate.
- $SE(I)$ is the standard error of I.
- z_A, z_B, z_C, z_D are the probit values of the four corners A, B, C, D.
- s_c^2 and n_c are the sample variance and sample size of corner c.

σ scales every term in Eq. (3) equally, so it cancels from the test. That is why the interaction test is the load-bearing result and the fold magnitudes are decoration.

Parameters

Symbol	Meaning	Value	Grade
p	positive fraction	measured, 4 corners × 3 replicates × 4 timepoints	**calibration**
θ	cytometry threshold	cancels — never needed	—
σ	cell-to-cell spread	**not identifiable from this dataset**	—
$\sigma_{9.23}$	spread reproducing last cycle's published folds	2.53	**not a measurement**
—	multiplicativity criterion	I within $2\,SE$ of zero	protocol

Substituting

At 3 h corner A reads 0 % positive, where Φ^{-1} diverges, so that timepoint cannot be inverted at all; our code raises rather than silently clipping, because clipping would fabricate a finite z.

24 h corner means (probit of each replicate, then averaged):

Corner	Positive (3 replicates)	$\bar{z}$	s
A 21 %	5.5 / 5.6 / 5.2 %	−1.6044	0.0190
B 1 %	31.9 / 31.3 / 31.9 %	−0.4761	0.0097
C 21 % + lactate	81.3 / 85.9 / 81.4 %	+0.9525	0.1068
D 1 % + lactate	98.3 / 98.9 / 96.4 %	+2.0699	0.2494

$$I_{24\,\mathrm{h}} = 2.0699 - 0.9525 - (-0.4761) + (-1.6044) = -0.0110$$

$$SE = \sqrt{0.0190^2 + 0.0097^2 + 0.1068^2 + 0.2494^2/3} = \sqrt{0.02469} = 0.1571$$

Reconstructing last cycle's published folds from Eq. (2) at $\sigma = 2.53$: $e^{2.53 \times 1.1283} = 17.37\times$ for oxygen against its published 17.4×, and 644.87× for lactate against 642× — the method was a probit inversion, and it implicitly chose that spread.

Results

Time	Oxygen $B - A$	Oxygen $D - C$	Lactate $C - A$	Lactate $D - B$	**Interaction**	t
6 h	0.726	1.010	0.823	1.107	**+0.284 ± 0.240**	1.18
9 h	0.592	1.727	1.170	2.306	**+1.135 ± 0.350**	**3.24**
24 h	**1.128**	**1.117**	**2.557**	**2.546**	**−0.011 ± 0.157**	−0.07

At 24 h the two oxygen estimates agree to 0.98 % and the two lactate estimates to 0.43 %, and the interaction is zero within error. The oxygen effect is the same with and without lactate, and the lactate effect the same under normoxia and hypoxia — which is what additivity, and so multiplicativity, means. These are two independent internal checks rather than one fit.

At 9 h the interaction is significantly positive ($t = 3.24$, 8 df, $p \approx 0.01$): the gate is *super*-multiplicative while it is still turning on. This is consistent with hypoxia acting on rate rather than level — three readouts put the 24 h hypoxia increment at only 1.18–1.33×, where mid-course it is 3.4×. A pure rate effect shows up as apparent synergy in the transient and disappears once both corners saturate.

So the AND claim belongs to the endpoint. Mid-course ON/OFF ratios must not be extrapolated with a multiplicative model. § 4.1 turns this from a test into an out-of-sample prediction.

The interaction test is the load-bearing result here; the fold magnitudes are decoration. σ cancels from the test (Eq. (3)) but not from a fold, so we report the folds as σ-conditional and build nothing on them:

σ	Oxygen fold	Lactate fold	Implied population CV
1.0	3.1×	12.9×	131 %
2.53	**17.4×**	**645×**	**2,452 %**
4.0	91.2×	27,660×	298,000 %

The last column is worth a line, because σ and the coefficient of variation are the same statement for log-normal expression ($CV = \sqrt{e^{\sigma^2} - 1}$). **The σ that reproduces last cycle's published 17× and 642× implies a population spread of about 2,450 %** — four decades of fluorescence. Possible for a strongly bimodal reporter, but it was never stated as a claim: it was fixed implicitly by inverting to the answer. We quote those two numbers only to say where they came from.

1.2 Lactate dominates the gate

Three readouts agree on how much the hypoxia input contributes at 24 h: positive fraction rises 1.18× (82.9 → 97.9 %), intensity 1.33× (253,571 → 337,332), and the untagged strain's endpoint OD_{600} not at all (1.380 → 1.380). Mid-course, at 6 h and 9 h, hypoxia multiplies the positive fraction by 3.4×.

Lactate is necessary and close to sufficient; hypoxia sets how fast the gate opens, not whether it opens. This is why § 2.3's luminal lactate load is close to the whole safety story.

1.3 Growth and carrying capacity

Question. Growth curves jump in the first hour and then creep, and no three-parameter sigmoid fits both phases. *Where* exactly is the misfit, is the first-hour burst real growth or a plate artefact, and does the misfit propagate into any conclusion we use?

Model

Two candidate laws, chosen so their *structures* differ diagnostically:

$$N(t) = K \exp\left(-b e^{-ct}\right) \qquad (5)$$

$$N(t) = K - (K - N_0) e^{-kt} \qquad (6)$$

Eq. (5) is the Gompertz form in the parameterisation standard for bacterial growth curves [5]. It has its maximum rate at an **interior inflection**; Eq. (6) has it at $t=0$. If the data's maximum rate is at $t=0$, the former *structurally* cannot fit it.

$$J=\frac{RMSE_{\text{with } t=0}}{RMSE_{t\geq 1\,\text{h}}} \qquad (7)$$

$$B=\frac{v_{0\to 1}(\text{induced})}{v_{0\to 1}(\text{uninduced})} \qquad (8)$$

$$K=\bar{y}_{24},\ \text{half-width}=t_{0.975,2}\frac{s}{\sqrt{n}},\ t_{0.975,2}=4.3027 \qquad (9)$$

where

- $N(t)$ is the bacterial density (OD_{600}) at time t.
- K is the carrying capacity, taken as the measured 24 h mean OD.
- N_0 is the initial inoculum density.
- b, c are the Gompertz shape parameters.
- k is the first-order approach rate (h^{-1}).
- J is the ratio of the fit error with the $t=0$ point included to the fit error from 1 h on.
- $RMSE$ is the root-mean-square error of the fitted growth curve.
- $v_{0\to 1}$ is the growth velocity over the first hour (OD h^{-1}).
- B is the first-hour growth-burst ratio, induced over uninduced.
- $\bar{y}_{24}$ is the mean 24 h OD across replicate measurements.
- s and n are the sample standard deviation and the number of replicate measurements.
- $t_{0.975,2}$ is the 97.5th percentile of Student's t distribution with 2 degrees of freedom (4.3027).

Substituting and results

Dropping the single point at $t=0$ and refitting the same Gompertz:

Construct	Condition	with $t=0$	**from 1 h**	J
GSDMD-SsrA	aerobic, no lactate	0.1469	**0.0262**	**5.61×**
GSDMD-SsrA	aerobic + lactate	0.1143	**0.0188**	**6.06×**
GSDMD	aerobic, no lactate	0.1628	**0.0221**	**7.37×**
GSDMD-SsrA	anaerobic, no lactate	0.0401	0.0410	0.98×

Construct	Condition	with $t=0$	**from 1 h**	J
GSDMD-SsrA	anaerobic + lactate	0.0405	0.0389	1.04×
GSDMD	anaerobic, no lactate	0.0292	0.0277	1.05×

The three normoxic corners improve 5.6–7.4×; the anaerobic corners do not move — they have no burst to misfit. Freeing the initial condition (Eq. (6)) does *not* help (0.1331 vs 0.1469, a 10 % gain, with fitted N_0 only 1.16–1.23), so the problem is the **shape** of the first hour, not the anchor value.

The burst is real growth. Same medium, same oxygen, different construct:

Construct	Condition	induced $v_{0\to1}$	uninduced	B
GSDMD-SsrA	aerobic + lactate	0.6367	0.8100	**0.786**
GSDMD	aerobic + lactate	0.1067	0.8267	**0.129**
GSDMD-SsrA	anaerobic + lactate	0.4767	0.4933	**0.966**
GSDMD	anaerobic + lactate	0.0800	0.4667	**0.171**

> **A plate artefact would not know which measurements came from which construct.** This one does: the tagged strain keeps 79–97 % of its first-hour burst and the self-toxic one keeps 13–17 %. The tagged construct serves as a control for the medium itself, ruling out "lactate slows growth".

This also sharpens the self-toxicity phenotype from "42 % less biomass at 24 h" to **"the first-hour growth burst is abolished by 83–87 %"**.

The misfit does not propagate. Every downstream use takes a *measured* 24 h value — site densities in § 2.1, the chassis-cost anchor in § 2.4 — and nothing in this project reads a value off a fitted curve. We assert this executably (`misfit_propagates() -> False`), so if any module ever starts using fitted trajectories the test suite fails.

What does propagate is the capacity itself, determined to **1.05 %–15.23 %** (tight normoxic, wider anaerobic). The tissue axis consumes *ratios* of these, so the precision is ample.

1.4 Construct selectivity

Pairing within a single experiment, the construct we build kills CT26 at 4.77 % and L929 at 1.43 % at 10 OD — a selectivity index of **3.3×** ($p<0.001$, $n=3$). Figures of 11.4× and 2.97× quoted earlier paired a numerator from one block with a denominator from another and are withdrawn (§ 5.1).

This is the one number on the page that nothing downstream depends on, and that is by design. 3.3× is modest, and an AND gate exists precisely so that it does not have to be large: the safety argument in chapter 2 is carried entirely by the circuit's *tissue* selectivity — 21.85× at the bottleneck from the sensors, before the effector contributes anything at all — rather than by any intrinsic preference of GSDMD-N for tumour cells. A design that needed the effector to be selective would be a worse design.

That matters for how the measurement should be read. The pair is a mouse colon carcinoma against a mouse fibroblast line, CT26.WT [6] and NCTC clone 929 / L-929 [7], which differ in tissue of origin as well as in malignancy. So 3.3× is "tumour line over a normal line", not "colon tumour over colon epithelium". We report it as the former and use it as neither: § 3.5's off-target argument is built on gate output and crypt renewal, not on this ratio.

---

2. Safety — where else does the gate switch on?

2.1 The tissue axis

> **Question.** With the gut lumen added to the site list, what is the gate output at each normal site relative to the tumour, and **which site is the bottleneck**? Is the "22–910×" range we inherited a measurement uncertainty or the span of one unidentified parameter?

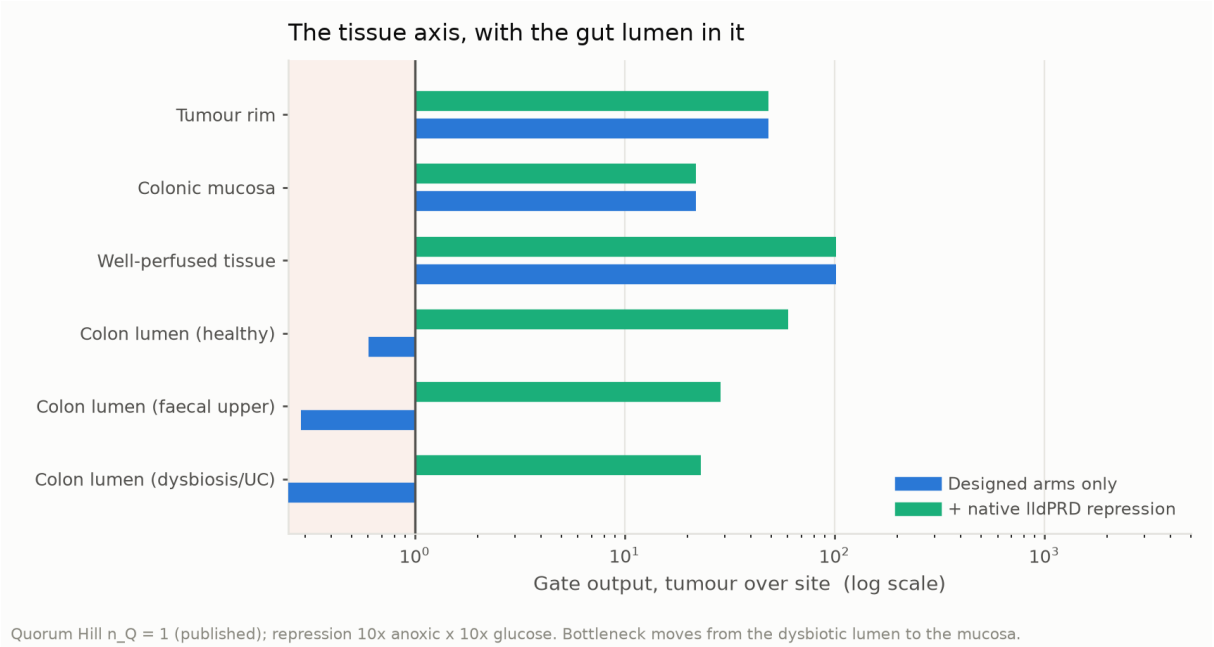


Figure 2. Gate output at each site relative to the tumour core (Eq. (11)), at the published $n_Q = 1$ and $K_{O_2} = 0.49$ %. Blue is the designed arms alone; green adds the native lldPRD repressions at 10× each. **Answers §2.1's question about the bottleneck:** on the designed arms alone all three luminal sites fall into the shaded region below 1.0, and it is the native repression — not the designed circuit — that moves the weakest link back to the colonic mucosa at 21.85×.

Model

Both designed sensors are Hill functions with a basal floor — the lactate arm *activated* by its stimulus, the hypoxia arm *repressed* by oxygen:

$$f_L(L) = b_L + (1 - b_L)\frac{L^{n_L}}{EC_{50}^{n_L} + L^{n_L}}, f_O(C) = b_O + (1 - b_O)\frac{K_{O_2}^{n_O}}{K_{O_2}^{n_O} + C^{n_O}} \qquad (10)$$

Because LuxI sits under pPepT, the quorum signal is proportional to hypoxia activity times density, $Q \propto f_O N$. Gate output at a site relative to the tumour core is therefore

$$R_{\text{margin}}(\text{site}) = \underbrace{\left(\frac{f_L(L_t)}{f_L(L_s)}\right)}_{\text{lactate arm}} \cdot \underbrace{\left(\frac{f_O(C_t)}{f_O(C_s)}\right)}_{\text{hypoxia arm}} \cdot \underbrace{\left(\frac{f_O(C_t)/f_O(C_s)}{N_s/N_t}\right)^{n_Q}}_{\text{quorum arm}} \cdot \underbrace{\left(\rho_{\text{anox}}\rho_{\text{glu}}\right)}_{\text{native repression}} \qquad (11)$$

Setting $R_{\text{margin}} = 1$ and solving for the luminal density ratio $d = N_s/N_t$ gives a critical density:

$$d_{\text{crit}} = \text{hyp} \cdot (\text{lac} \cdot \text{hyp} \cdot R_{\text{rep}})^{1/n_Q}, R_{\text{rep}} = \rho_{\text{anox}}^{-1}\rho_{\text{glu}}^{-1} \qquad (12)$$

where

- $f_L(L)$ is the lactate-arm activity, a Hill function of lactate with a basal floor (Eq. (10)).

- $f_O(C)$ is the hypoxia-arm activity, a Hill function repressed by oxygen (Eq. (10)).
- L and C are the lactate concentration (mM) and the oxygen concentration (% O_2).
- b_L, n_L and EC_{50} are the lactate-arm basal fraction, Hill coefficient and half-maximal lactate concentration (1/7, 1.0, 1.6 mM).
- b_O, n_O and K_{O_2} are the hypoxia-arm basal fraction, Hill coefficient and half-maximal repression oxygen (0.05, 2.0, 0.49 % O_2).
- Q is the quorum signal, proportional to hypoxia activity times bacterial density.
- N is the bacterial density (relative OD), assigned from the measured capacities of § 1.3.
- $R_{\text{margin}}(\text{site})$ is the gate output at a site relative to the tumour core.
- n_Q is the quorum Hill coefficient (1.0 published, swept 0.95–1.5).
- ρ_{anox} and ρ_{glu} are the residual activities of the native anoxic and glucose repressions (0.1 each).
- $d = N_s/N_t$ is the luminal density ratio, and d_{crit} its critical value at which $R_{\text{margin}} = 1$.
- $\text{hyp} = f_O(C_t)/f_O(C_s)$ and $\text{lac} = f_L(L_t)/f_L(L_s)$ are the hypoxia- and lactate-arm ratios.
- $R_{\text{rep}} = \rho_{\text{anox}}^{-1}\rho_{\text{glu}}^{-1}$ is the native-repression product.
- Subscripts t and s denote the tumour core and the site.

Note the scaling: **the margin repression buys grows only as** R_{rep}^{1/n_Q}, so a strong repression is worth less when the quorum arm is steep.

The quorum arm is treated as sub-threshold, i.e. the tumour sits *below* the quorum set point. This is inherited rather than invented: if the tumour were saturated the arm would contribute no selectivity at all, and neither the tissue folds nor the lesion-size turn-on of § 2.2 could arise.

Parameters

Symbol	Value	Unit	Grade	Note
EC_{50}, n_L, b_L	1.6, 1.0, 1/7	mM, —, —	scenario	measured wild-type lldPRD [8, 9]
K_{O_2}, n_O, b_O	**0.49**, 2.0, 0.05	% O_2, —, —	**literature prior** / scenario	K_{O_2} from [10]; 9.23 used 1.0 — see § 2.5
n_Q	**1.0** (swept	—	**literature**	published

Symbol	Value	Unit	Grade	Note
	0.95–1.5)		**prior**	pLux fits [11]; 9.23 used 2.0 — see § 2.5
N	measured capacities below	rel. OD	**calibration**	§ 1.3
ρ_{anox}, ρ_{glu}	0.1 each	—	scenario	placeholder; § 2.3
d	1.0 (swept 0.1–10)	—	**unmeasured**	luminal EcN density

Bacterial density is **measured, not assumed**: each site is assigned the carrying capacity its own oxygen and lactate condition produced in our growth block — aerobic/no-lactate 2.6133, aerobic/lactate 2.3600, anaerobic/no-lactate 1.5667, anaerobic/lactate 1.5067. Every site is evaluated at its own capacity, which assumes EcN colonises healthy tissue as well as it colonises tumour and is the conservative choice.

Substituting

With $K_{O_2}=1$ %, $n_O=2$, $b_O=0.05$, Eq. (10) becomes $f_O(C)=0.05+0.95\,K_{O_2}^2/(K_{O_2}^2+C^2)$ with $K_{O_2}=0.49$ %, i.e. $K_{O_2}^2=0.2401$:

Site	C	$f_O(C)$	hypoxia arm
Tumour core (target)	0.5 %	$0.05+0.95\cdot 0.2401/0.4$	—
Tumour rim	5 %	0.05904	8.730
Colonic mucosa	2 %	0.1038	4.966
Well-perfused tissue	10 %	0.05228	9.859
Colon lumen (all)	**0 %**	$0.05+0.95=1.0000$	**0.5154** ❌ **< 1**

Colonic mucosa, continuing through Eq. (11) with $n_Q=1$:

$$d_{\text{mucosa}}=\frac{2.6133}{1.5067}=1.7345\,,\text{quorum}=\left(\frac{4.9656}{1.7345}\right)^1=2.8628$$

$$R_{\text{mucosa}}=1.5372\times 4.9656\times 2.8628\times 1=21.85\times$$

Results

At the published sensor values, **without** native repression:

Site	Lactate arm	Hypoxia arm	Density ratio	Quorum arm	**Margin** R_{margin}	Protective?
Tumour rim	1.0000	8.730	1.5664	5.574	**48.66×**	✅
Colonic mucosa	1.5372	4.966	1.7345	2.863	**21.85×**	✅
Well-perfused tissue	1.8140	9.859	1.7345	5.684	**101.7×**	✅
Colon lumen (healthy)	2.2759	**0.5154**	1.0000	0.5154	**0.605×**	❌
Colon lumen (faecal)	1.0820	**0.5154**	1.0000	0.5154	**0.287×**	❌
Colon lumen (dysbiotic)	0.8689	**0.5154**	1.0000	0.5154	**0.231×**	❌

Adding the native repressions (100× combined, acting only at the anoxic glucose-rich luminal sites) moves the healthy lumen to 60.5×, faecal to 28.7× and dysbiotic to 23.1×, while leaving the mucosa at 21.85× untouched. **So the native repression does not merely help the lumen — it changes which site is the weakest link:**

	Bottleneck	Margin
Without repression	**dysbiotic lumen**	**0.231× (unsafe)**
With repression	**colonic mucosa**	**21.85× (safe)**

Note that all three luminal sites are now *unambiguously* below 1.0 without the native repression, where the inherited priors left the healthy lumen marginal at 1.21×. The published oxygen affinity makes the designed circuit's luminal problem **worse**, and the native repression correspondingly more load-bearing.

The "22–910×" range is a span of n_Q, not a measurement uncertainty. Last cycle's two calibrations shared every fixed prior and differed almost entirely in that one coefficient (n_Q=1.000 and 2.978). Sweeping it, now against the published value rather than last cycle's:

n_Q	Colonic mucosa	Well-perfused
0.95	20.7×	93.2×
1.0 (published centre)	**21.9×**	**101.7×**
1.5 (published upper)	37.0×	242×
2.0 (9.23’s prior)	62.6×	578×
3.0	179×	3,285×
4.0	513×	18,671×

An exogenous-AHL dose response on pLux would pin n_Q down in an afternoon.

> ⚠ **Honest caveat.** The reconstructed span over $n_Q \in [1,4]$ (mucosa 10.1–74.4×) does **not** cover the published 22.3–909.6×. This model is a homogeneous point estimate where last cycle solved a spatially averaged PDE, so we claim agreement in direction and driver, **not** reproduction.

Last cycle’s lactate priors were considerably more generous than the published characterisation of the part we actually chose (EC_{50} 5.0 vs 1.6 mM, n_L 2.0 vs 1.0, basal 0.0078 vs $\approx 1/7$ — all three shifted toward better discrimination, none anchored to BBa_M36021). That inflated the lactate arm by 5.23× at the mucosa and 14.48× at the healthy lumen. But at the dysbiotic lumen the inflation factor is **0.83× — the generous prior is actually worse**, because at 100 mM both priors saturate. **Safety cannot be bought by changing priors; the correction matters because it stops us telling a flattering story, not because it changes the verdict.**

The one remaining free parameter is luminal EcN density. Eq. (12) gives $d_{\text{crit}} = 0.5154 \times (0.8689 \times 0.5154 \times 100)^{1/1} = 23.08$ with repression, against 0.231 without: **the lumen stays off as long as luminal EcN density does not exceed roughly 23× the tumour’s.** At $n_Q = 1$ the critical density scales linearly in the repression fold rather than as its square root, which is why this threshold is three times what the inherited prior gave.

There is one human measurement bearing on that threshold, and it is a warning as much as a reassurance. In a double-blind trial, patients took 10^9 CFU of EcN daily for 14 days before resection; tumour tissue was enriched for EcN over matched normal tissue ($p = 0.03$), **but two of the six treated participants carried comparably high EcN in adjacent normal colon and showed no tumour enrichment at all** [1]. Six patients cannot give a distribution for d, and we do not treat them as one. What they do establish is that $d > 1$ happens in people — so the

sweep to $d=10$ is covering a real case, not a hypothetical one, and the 23.08 threshold is the number to watch.

2.2 Lesion size: a floor that needs no calibration

> **Question.** Below what radius can the hypoxia arm not leave basal **for any parameter values**? And is the inherited "14×–6891×" size turn-on the same uncertainty as §2.1's, or independent evidence?

Model

Oxygen penetrates a fixed depth d_{O_2} from the perfused surface, so

$$r_{\text{core}}(R)=max(R-d_{O_2},0),\phi_{\text{hyp}}(R)=\left(\frac{r_{\text{core}}(R)}{R}\right)^3 \qquad (13)$$

$$\bar{f}_O(R)=\phi_{\text{hyp}}f_O(C_{\text{core}})+(1-\phi_{\text{hyp}})f_O(C_{\text{rim}}) \qquad (14)$$

$$G(R)=\left[\bar{f}_O(R)\cdot\Psi_{\text{centre}}(R)\right]^{n_Q} \qquad (15)$$

where Ψ_{centre} is the closed-form AHL volume-integration kernel of §3.2 (Eq. (35)) — no PDE solve is needed. The two mechanisms separate cleanly *before* the Hill exponent is applied:

$$\underbrace{\left(\frac{\bar{f}_O(R_2)}{\bar{f}_O(R_1)}\right)}_{\text{hypoxic volume}}\times\underbrace{\left(\frac{\Psi_{\text{centre}}(R_2)}{\Psi_{\text{centre}}(R_1)}\right)}_{\text{AHL integration}}=\frac{Q(R_2)}{Q(R_1)} \qquad (16)$$

And a direct corollary of Eq. (13), with no sensor parameter in it at all:

$$R<d_{O_2}\Rightarrow\phi_{\text{hyp}}\equiv 0\Rightarrow\bar{f}_O\equiv f_O(C_{\text{rim}})=\text{const} \qquad (17)$$

where

- R is the lesion radius (μm).
- d_{O_2} is the oxygen penetration depth from the perfused surface (178 μm, from the 1 % O_2 contour at 222 μm in a 400 μm nodule).
- $r_{\text{core}}(R)=max(R-d_{O_2},0)$ is the hypoxic-core radius.
- $\phi_{\text{hyp}}(R)=(r_{\text{core}}/R)^3$ is the hypoxic volume fraction.
- $f_O(C)$ is the hypoxia-arm Hill function of Eq. (10).
- $\bar{f}_O(R)$ is the volume-averaged hypoxia-arm activity.
- C_{core} and C_{rim} are the oxygen concentrations at the lesion core and rim (0.5 % and 5 %).
- $\Psi_{\text{centre}}(R)$ is the normalised AHL at the lesion centre (Eq. (35)).
- n_Q is the quorum Hill coefficient.
- $Q(R)$ is the quorum signal for a lesion of radius R.
- $G(R)$ is the gate output for a lesion of radius R.

Substituting

d_{O_2} is calibrated to last cycle's own statement that the 1 % O_2 contour sits at 222 µm in a 400 µm nodule, i.e. $400-222=178\,\mu m$.

At $R=150$ µm: $r_{\text{core}}=0$, so $\bar{f}_O=f_O(5\%)=0.059037$, independent of R. At $R=400$ µm: $\phi_{\text{hyp}}=(222/400)^3=0.170954$ and

$$\bar{f}_O(400)=0.170954\times0.515405+0.829046\times0.059037=0.137055$$

Results

> **A lesion below ≈178 µm has no hypoxic volume whatsoever, so the hypoxia arm cannot leave basal however the circuit is parameterised.** This is pure geometry — zero fitted parameters — and it is the hardest safety statement in this work.

The floor agrees with a seventy-year-old histological measurement

Our 178 µm comes from last cycle's own oxygen contour, with no histology in it. It is worth checking against the measurement the whole field's diffusion distance comes from. Thomlinson and Gray sectioned human lung carcinomas and found that **the radius of intact tumour cells surrounding a necrotic core never exceeded 180 µm**; central necrosis was *absent* in every section below 160 µm radius and *present* in every section above 200 µm [12, 13].

That is a measured bracket for exactly the quantity d_{O_2} represents, and our independently derived value falls inside it, 1 % below its upper edge:

d_{O_2}	Source	Floor	$\phi_{\text{hyp}}(400\,\mu m)$	150→1000 µm turn-on
160 µm	histology, necrosis absent below	160 µm	0.2160	2.07×10^{-5}
178 µm	**our oxygen contour**	**178 µm**	**0.1710**	2.31×10^{-5}
200 µm	histology, necrosis present above	200 µm	0.1250	2.63×10^{-5}

The floor *is* d_{O_2} by construction, so the histological bracket transfers to it directly: **the smallest lesion that can develop a hypoxic core is 160–200 µm, and 178 µm is our reading inside that range.** Across the whole bracket the 150→1000 µm turn-on moves by 27 %, which changes nothing — the statement being made is that small lesions have *no* hypoxic volume, and where that threshold sits is bounded by histology rather than by our circuit.

This is the only parameter in the project where an independent measurement lands on a value we derived from somewhere else entirely. We flag it because the reverse — a derived number that the literature contradicts — is what § 2.5 found for the oxygen affinity, and a reader is entitled to know which is which.

Decomposing the 150 → 1000 µm turn-on by Eq. (16):

Mechanism	Fold
Hypoxic volume (mean pPepT activity)	**5.293×**
AHL volume integration	**36.907×**
Combined quorum signal	**195.4×**

The AHL term carries 6.5× the weight of the hypoxia term — the size switch is fundamentally a *volume integrator*, which § 3.2 reaches independently.

Applying only n_Q to last cycle's own signal fold of 22× reproduces its published range: $22^{1.000}=22.0$ against 14.1, and $22^{2.978}=9948$ against 6891 — both within 1.6× across three orders of magnitude.

> **The tissue-axis range (22–910×) and the size range (14–6891×) are the same unidentified Hill coefficient propagating through a power law. They must not be presented as two independent pieces of evidence.**

⚠ Our free-field kernel has no boundary sink, giving a signal fold of 195.4× against last cycle's 22× — an overestimate of **9.47×**, because its PDE imposed a Robin washout at the nodule surface that drains a large nodule's integration advantage. Use 22× when an absolute value is needed; use this section for the mechanism split and the geometric floor.

2.3 The colon lumen

> **Question.** We dose orally, so the bacteria sit in the colon lumen first. Are the two designed arms *anti*-selective there — and can that be decided without the sensor parameters we cannot measure? If they are, what holds the lumen off, and how strong does it need to be?

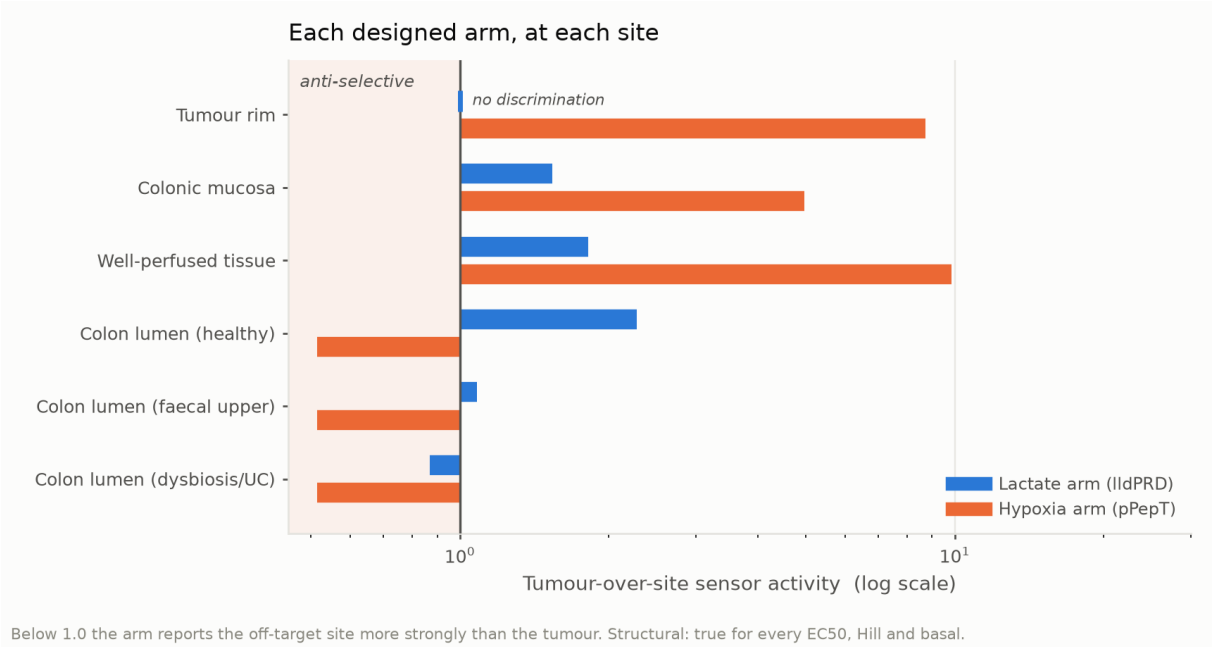


Figure 3. Each designed arm on its own, tumour over site. Anything left of 1.0 reports the off-target site *more* strongly than the tumour. **Answers §2.3's first question:** the hypoxia arm is anti-selective in all three luminal scenarios, because the lumen is anoxic where the tumour core is only hypoxic, and under dysbiotic lactate the lactate arm inverts too. By Eq. (18) this holds for every EC_{50}, Hill coefficient and basal level, so no amount of tuning repairs it.

Model

The lactate sensor's EC_{50} and Hill coefficient are **not identifiable** from our data: the reporter experiment has two oxygen levels and two lactate levels, which locate neither a midpoint nor a slope, and the lactate gradient that would fix them will not be run this cycle. So we invert the question — *for which sensor parameters is the lumen held off while the tumour is on?* — and find that for both designed arms the parameters are not needed at all.

Both arms are monotone in their stimulus (Eq. (10)). Therefore

$$s_{\text{offsite}} \geq s_{\text{target}} \Rightarrow \frac{f(s_{\text{target}})}{f(s_{\text{offsite}})} \leq 1 \text{ for every } (EC_{50}, n, b) \qquad (18)$$

with hypoxia stimulus expressed monotonically as $s_O = -C$.

The lactate arm also has an analytic ceiling. Pushing EC_{50} far above both concentrations puts them on the initial power-law stretch $f_L \sim L^{n_L}$, so the ratio approaches $(L_t/L_{\text{off}})^{n_L}$ **from below and never exceeds it**; a non-zero basal level caps it again at $1/b_L$:

$$\underset{EC_{50}}{¿} \frac{f_L(L_t)}{f_L(L_{\text{off}})} = min\left\{\left(\frac{L_t}{L_{\text{off}}}\right)^{n_L}, \frac{1}{b_L}\right\} \qquad (19)$$

$$n_L \geq \frac{\ln R_{\text{req}}}{\ln\left(L_t / L_{\text{off}}\right)} \qquad (20)$$

The native repressions enter multiplicatively on the luminal side only:

$$R_{\text{margin}} = \frac{f_L\left(L_t\right)}{f_L\left(L_{\text{off}}\right)\rho_{\text{anox}}\rho_{\text{glu}}}, \left(\rho_{\text{anox}}\rho_{\text{glu}}\right)^{-1} \geq \frac{R_{\text{req}}}{R_{\text{derepressed}}} \qquad (21)$$

The one candidate among the designed inputs that is *not* structurally anti-selective is luminal washout:

$$\eta_{\text{ret}} = \frac{k_{\text{decay}}}{k_{\text{decay}} + k_{\text{wash}}}, k_{\text{decay}} = \frac{\ln 2}{t_{1/2}^{\text{AHL}}}, k_{\text{wash}} = \frac{1}{\tau_{\text{transit}}} \qquad (22)$$

where

- s_{target} and s_{offsite} are the stimulus strength at the tumour and at an off-target site; for the hypoxia arm the stimulus is $s_O = -C$.
- $f(s)$ is the sensor activity as a function of its stimulus.
- EC_{50}, n and b are the sensor half-maximal stimulus, Hill coefficient and basal fraction.
- L_t is the tumour lactate concentration (8.0 mM).
- L_{off} is the off-target luminal lactate concentration (0.6–100 mM across the luminal scenarios).
- n_L and b_L are the lactate-arm Hill coefficient and basal fraction (1.0, 1/7).
- R_{req} is the required margin (10× here).
- $R_{\text{derepressed}}$ is the margin obtained without the native repressions.
- ρ_{anox} and ρ_{glu} are the residual activities of the native anoxic and glucose repressions, entering multiplicatively on the luminal side only.
- η_{ret} is the fraction of AHL retained in the lumen.
- $k_{\text{decay}} = \ln 2 / t_{1/2}^{\text{AHL}}$ is the AHL first-order decay rate, and $t_{1/2}^{\text{AHL}}$ the AHL half-life (8 h).
- $k_{\text{wash}} = 1/\tau_{\text{transit}}$ is the luminal washout rate, and τ_{transit} the colonic transit time (swept 1–24 h).

Parameters

Symbol	Value	Grade	Source
L_t	8.0 mM	literature prior	the median of the clinical series, ≈8 μmol g^{-1} [14]
L_{off} healthy	0.6 mM	literature prior	rectal lumen by equilibrium dialysis, L-lactate

Symbol	Value	Grade	Source
			0.5 (0–1.6), D-lactate 0.1 (0–0.8) mM, $n=11$ healthy controls [15]
L_{off} faecal	5.0 mM	literature prior	lactate < 5 mM in healthy faeces [16]
L_{off} dysbiotic	100 mM	**scenario only**	faecal lactate to 110 mM in short bowel [17], raised in active UC [18]
C_{off}	**0 %**	binary flag	lumen anoxic — see § 2.3.4
$t_{1/2}^{AHL}$	8 h	scenario	placeholder
$\tau_{transit}$	swept 1–24 h	scenario	colonic transit is slow (24–72 h)

Substituting

With $n_L=1$, $EC_{50}=1.6$ mM, $b_L=1/7$:

$$f_L(8.0)=1/7+6/7\cdot\frac{8.0}{9.6}=0.857143, f_L(0.6)=1/7+6/7\cdot\frac{0.6}{2.2}=0.376623$$

giving 2.276× at the healthy lumen; likewise 1.082× at faecal lactate and **0.869× — already inverted** — in dysbiosis.

Comparing stimuli site by site (Eq. (18)):

Site	O_2	Lactate	Lactate arm	Hypoxia arm
Tumour core (target)	0.5 %	8 mM	—	—
Colonic mucosa	2 %	1.5 mM	✅ 1.54×	✅ 3.38×
Well-perfused	10 %	1 mM	✅ 1.81×	✅ 13.63×
Colon lumen (healthy)	**0 %**	0.6 mM	✅ 2.28×	❌ **anti-selective**
Colon lumen (faecal)	**0 %**	5 mM	⚠ 1.08×	❌ **anti-selective**
Colon lumen (dysbiotic)	**0 %**	**100 mM**	❌ **anti-selective**	❌ **anti-selective**

Results

> **The colon lumen is anoxic (0 % O_2) where the tumour core is merely hypoxic (0.5 %), so the hypoxia arm reads the lumen *more strongly* than the tumour in all three luminal scenarios. In dysbiosis both designed arms invert simultaneously, and by Eq. (18) no parameter choice whatsoever can repair this.**

This bites harder than it looks, because § 1.2 established that the gate is **lactate-dominated** — the arm that actually does the work is the one that fails under high luminal lactate.

The ceiling (Eq. (19)) says where the limit comes from:

Site	L_{off}	Wild type	$n_L=1$	$n_L=2$	$n_L=4$	n_L needed for 10×
Lumen (healthy)	0.6 mM	2.28×	13.3×	178×	31,605×	**0.89**
Lumen (faecal)	5 mM	1.08×	1.60×	2.56×	6.55×	**4.90**
Lumen (dysbiotic)	100 mM	0.87×	1.00×	1.00×	1.00×	**impossible**

Those shape bounds are then **truncated by the basal term**: with wild-type $b_L=1/7$, $min\{178,7\}=7.0\times$.

> **It is the concentration ratio between the two sites and the basal level — not the sensor's midpoint — that set what is achievable.** Wild-type lldPRD is capped at 7× however it is tuned, and reaching 10× against faecal lactate would need $n_L \geq 4.9$ while LldR has no reported cooperativity.

What holds the lumen off is the part we chose. BBa_M36021 is the native lldPRD promoter, and it retains the native anoxic and glucose repressions that engineered variants were built to remove. The two environments differ in exactly those two respects:

	Colon lumen	Tumour core
Oxygen	**anoxic (0 %)**	hypoxic but **not anoxic**
Glucose	dietary, proximally **rich**	**depleted** (glycolysis)

The repressions engage where we need them and relax where we do not. With 0.1 residual activity each (100× combined):

Site	Native (repressed)	Engineered (de-repressed)	Repression needed for 10×
Lumen (healthy)	**227.6×**	2.28×	⩾ 4.39×
Lumen (faecal)	**108.2×**	1.08×	⩾ 9.24×
Lumen (dysbiotic)	**86.9×**	0.87×	⩾ 11.51×

The worst case asks for about 11.5× combined, and the published characterisation of this promoter reports both repressions as complete.

That characterisation is the one we take EC_{50} and the dynamic range from, so it is not a separate assumption. Rebuilding the wild-type lldPRD biosensor on a low-copy plasmid, the authors found **no detectable response in the presence of 0.4 % glucose**, and **no response at all after 16 h under anaerobic conditions** [8] — the whole point of their engineering effort was to remove those two repressions, and the resulting ALPaGA promoter is a *different part* from the one we built. The native repressions are the documented behaviour of BBa_M36021 under exactly the two conditions the colon lumen imposes: glucose-rich and anoxic. LldR's control of the operon, and its loss of induction in an *lldR* deletion, are established separately [19].

We nevertheless credit each repression with only **10×**. A promoter that is "not detectably induced" in an assay with a finite floor could be repressed 10× or 10,000×, and the margin scales as R_{rep}^{1/n_Q} (Eq. (12)), so the difference is worth at most a factor of 10 in the margin — but nothing below is allowed to depend on the generous reading.

Washout (Eq. (22)) contributes little: with $t_{1/2}^{\mathrm{AHL}}$=8 h, suppression is 12.5× at 1 h transit but only **1.48× at 24 h** — and colonic transit is slow, landing at the weak end. It is a secondary mechanism.

That 8 h is a **declared placeholder, not a measurement**, and we looked for one. The standard study of AHL lactonolysis reports only *relative* rates — on a scale where C8-HSL is 1, our signal 3-oxo-C6-HSL is 170 at 37 °C — together with the finding that the 3-oxo substitution destabilises the ring and that warming from 22 to 37 °C speeds its opening 2.4× [20]. No absolute half-life at pH 7 is given, so none can be cited. What the paper does establish is the *direction*: 3-oxo-C6-HSL sits near the unstable end of the AHL series, and the large intestine's pH of 6.5–7.5 (the paper's own table of body fluids) is in the range where hydrolysis proceeds. A shorter true half-life means **more** luminal decay and a **larger** safety margin, so the placeholder errs against us, and this section claims nothing from it either way.

🔴 **Design consequence: do not "fix" this promoter.** Swapping in a de-repressed variant to raise tumour expression would discard the only mechanism holding the lumen off — discrimination falls from 86.9–227.6× back to 0.87–2.28×, and the full gate from 23–60× to 0.23–0.60×. That is the difference between a margin and an inversion.

2.3.4 The one assumption that cannot fail quietly

Every other open parameter in this work widens an interval. This one **inverts a conclusion**, and it is a binary with no measurement behind it:

Assumption	the colon lumen is anoxic (0 % O_2) and the tumour core (≈0.5 %) is not
Form	**binary flag, not a fittable parameter**
Measured half	at 1 % O_2 with lactate the circuit reaches its highest corner, 97.9 % positive ⇒ the promoter is plainly **not** repressed at 1 %, which is what the tumour side needs
Untested half	nothing in our dataset is at 0 % O_2
Failure mode	**sign change, not a wider interval**

Two things argue it holds: the requirement is modest (4.39× healthy, 9.24× faecal, 11.51× dysbiotic) against two repressions each reported as strong, and FNR-type oxygen regulation switches over exactly the 0–1 % range that separates the two sites. **But modest and reported is not measured, and we would rather say so than let it pass as established.**

One plate settles it: the reporter at 0 % against 1 % O_2, lldPRD-only.

2.4 SsrA as a post-translational valve

Question. Keep the degron or drop it — and what is the criterion? What is the effector's half-life in normal tissue? Is a second degron warranted? And is SsrA a *sufficient* safety valve or only an auxiliary one?

Model

Delivered exposure is biomass × per-cell potency, so dropping the tag multiplies potency by r_{potency} and biomass by r_{biomass}:

$$\frac{\text{delivery, tagged}}{\text{delivery, untagged}} = \frac{1}{r_{\text{biomass}}\, r_{\text{potency}}} > 1 \Leftrightarrow r_{\text{potency}} < r^{\dagger} \equiv \frac{1}{r_{\text{biomass}}} \qquad (23)$$

Keep the tag if and only if the potency it costs is less than the biomass it buys. This uses two measured ratios and assumes nothing about dose–response shape.

Treating degradation strength as a continuous knob:

$$P(\delta)=\frac{\delta_{\text{nat}}+\delta_{\text{built}}}{\delta_{\text{nat}}+\delta},\ \text{potency}=\frac{P}{k_{\text{export}}+P},\ \text{viability}=e^{-\lambda P} \qquad (24)$$

Each of the latter two carries exactly **one** free parameter, matched to the two measured states — the identifiability ceiling of this dataset.

$$t_{1/2}=\frac{\ln 2}{\delta_{\text{nat}}+\delta_{\text{tag}}} \qquad (25)$$

Finally, ClpXP is a finite enzyme. Michaelis–Menten degradation gives an *effective* first-order rate that **falls** as the pool grows:

$$v=\frac{v_{max}P}{K_m+P},\ k_{\text{eff}}(P)=\frac{v_{max}}{K_m+P} \qquad (26)$$

so amplification exists only inside a capacity window:

$$\text{leak rate} < v_{max} < \text{induced rate} \qquad (27)$$

where

- r_{biomass} is the 24 h biomass ratio of the untagged strain to the tagged (1.380/2.360 = 0.585).
- r_{potency} is the per-cell potency ratio of the untagged strain to the tagged.
- $r^{¿}=1/r_{\text{biomass}}$ (1.71) is the decision threshold; keep the tag iff $r_{\text{potency}}<r^{¿}$.
- $P(\delta)$ is the normalised steady-state effector pool.
- δ is the total effector degradation rate; δ_{nat} (0.08 h^{-1}) and δ_{tag} (0.43 h^{-1} as built) are its natural and tag-added components.
- $\text{potency}=P/(k_{\text{export}}+P)$ is the per-cell killing potency, and k_{export} its half-saturation constant.
- $\text{viability}=e^{-\lambda P}$ is the chassis survival fraction, and λ the viability penalty coefficient.
- $t_{1/2}=\ln 2/(\delta_{\text{nat}}+\delta_{\text{tag}})$ is the effector half-life.
- v is the ClpXP degradation rate; v_{max} and K_m are its maximum rate and Michaelis constant.
- $k_{\text{eff}}(P)=v_{max}/(K_m+P)$ is the effective first-order degradation rate at pool P.
- leak rate and induced rate are the promoter leak and fully induced synthesis rates; amplification exists iff leak rate ¿ v_{max} < ¿ induced rate.

Substituting

$$r_{\text{biomass}} = \frac{1.380}{2.360} = 0.58475 \Rightarrow r^{¿} = 1.71014$$

Dose	untagged	tagged	n	r_{potency}	Margin $r^{¿}/r$	Verdict
6 OD	2.900 %	3.267 %	3	**0.8878**	**1.926×**	✅ keep
8 OD	7.150 %	6.733 %	2 ⚠	**1.0619**	**1.610×**	✅ keep
10 OD	22.333 %	15.800 %	3	**1.4135**	**1.210×**	✅ keep

Calibrating Eq. (24)/4 on the most conservative anchor gives $\lambda = 0.12265$, $k_{\text{export}} = 0.56104$. Untagged pool from Eq. (24): $P(0) = 0.43/0.08 = 5.375$. Delivery at the built strength $\delta = 0.35$: $P = 1.000$, potency ¿ $1/1.56104 = 0.6406$, viability ¿ $e^{-0.12265} = 0.8846$, so 0.5667; untagged gives 0.4685. Their ratio **1.2099×** matches Eq. (23)'s $1/(0.58475 \times 1.4135)$ exactly — two independent routes agreeing.

Results

Keep the tag. All three doses satisfy $r < r^{¿}$, with margins 1.21–1.93×. At the lowest dose $r < 1$: the tagged construct actually kills slightly *more*. The 3.3× spread in potency cost we inherited was an artefact of pairing a numerator from one block with a denominator from another; the only legitimate value is 1.4135, so what was recorded as an open question is in fact settled. Had the rejected 4.6993 been used, Eq. (23) would give 0.364× and **invert** the decision.

Effector half-life in normal tissue (Eq. (25)):

Variant	δ_{total}	**Half-life**
Untagged	0.08 h^{-1}	**8.66 h**
SsrA as built	0.43 h^{-1}	**1.61 h**
Double SsrA (hypothetical)	0.78 h^{-1}	0.89 h

Both figures are declared rather than measured on our construct, so it is worth saying where they sit. The SsrA system's half-lives were characterised by appending the tag's last residues to GFP in *E. coli*: the variants span **about 40 minutes to a few hours**, set by the final three residues [21]. Our tagged value, 1.61 h (97 min), is inside that span, and the untagged 8.66 h is outside it on the stable side — which is what an untagged protein should be. Neither number is a measurement of GSDMD-N, and § 2.4's conclusion does not need one: the decision is a *ratio* comparison (Eq. (23)), and § 4.4 refits the exponent per construct rather than transporting a rate.

A second degron is not warranted. The tag as built is already 2.35× stronger than the delivery optimum (δ^{*}=0.149 h^{-1}), at a cost of **7.34 %** in delivered exposure — far below the cost of rebuilding and revalidating, and § 2.4's amplification argument says over-strength helps on the safety side.

> **SsrA is a genuine safety valve, not an auxiliary — because ClpXP saturates.** By Eq. (26), degradation is strong where the only substrate is promoter leak ($P \ll K_m$) and is outrun where the gate is open ($P \gg K_m$). **The protein-level tumour/normal ratio therefore exceeds the promoter's transcription ratio: the degron amplifies selectivity rather than merely damping output.** The same mechanism explains why over-strength is not a defect — a stronger tag pushes normal tissue deeper into the unsaturated regime.

⚠ v_{max} and K_m for EcN ClpXP on this substrate are unknown, so the amplification factors we can compute (peaking near 70.9× with demonstration parameters) are **illustrative, not predictive**. What this section reports is the *structural requirement* (Eq. (27)) — the same methodology as § 2.3: when a parameter cannot be pinned down, state where it must lie.

A separate finding, from § 3.1: **about 88.5 % of the degron's spatial safety benefit comes from making less, not from confining what is made.**

2.5 Which parameters decide the margin

> **Question.** Across the declared ranges, which parameters dominate the variance of the *bottleneck* margin — and are they measurable?

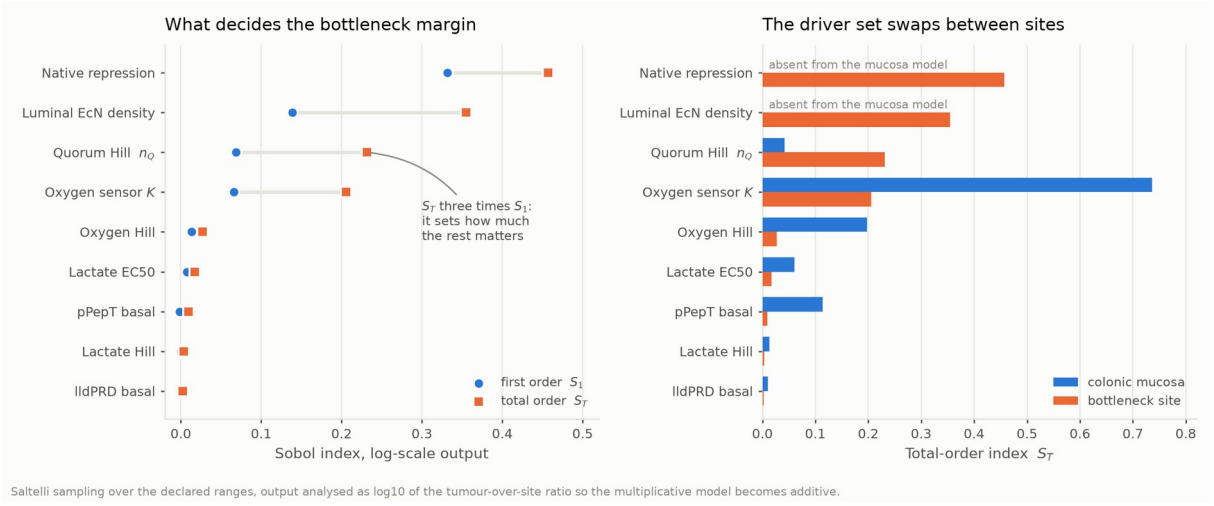


Figure 4. Left: first- and total-order Sobol indices for the bottleneck margin, computed on $\log_{10}$ of the ratio so that the multiplicative model becomes additive (Eq. (28)). **The gap between the two markers is the quantity to read** — n_Q's total order is 3.4× its first order, which is what it means for a parameter to set how much the others matter. Right: the same total-order index for the colonic mucosa against the bottleneck. **Answers § 2.5:** the two driver sets barely overlap, and the

two parameters that dominate the bottleneck do not exist at all in a mucosa-only model.

Model

The tissue ratio is a product, so we analyse on a **log scale**, which turns Eq. (11) into a sum:

$$\log R_{\text{margin}} = \log \text{lac} + \log \text{hyp} + n_Q \log \frac{\text{hyp}}{d} + \log R_{\text{rep}} \qquad (28)$$

A purely multiplicative model would then show **zero interaction** and first-order indices summing to 1. That makes the analysis *diagnostic* rather than merely descriptive: anything that survives is a real coupling. Eq. (28) contains exactly one — n_Q multiplies $\log(\text{hyp}/d)$ — so we can **predict in advance** that n_Q will carry a modest first-order index and a much larger total-order index.

$$S_i = \frac{V_i}{Var\left[\log R_{\text{margin}}\right]}, S_{Ti} = 1 - \frac{V_{\sim i}}{Var\left[\log R_{\text{margin}}\right]} \qquad (29)$$

The gap between the two is what the parameter does *through* the others, so the share of a parameter's influence that is interaction is

$$\text{interaction share}_i = 1 - \frac{S_i}{S_{Ti}} \qquad (30)$$

where

- R_{margin} is the bottleneck margin, the gate output at the limiting site.
- lac, hyp and d are the lactate arm, hypoxia arm and luminal density ratio of Eq. (11).
- n_Q is the quorum Hill coefficient.
- $R_{\text{rep}} = \rho_{\text{anox}}^{-1} \rho_{\text{glu}}^{-1}$ is the native-repression product.
- S_i and S_{Ti} are the first- and total-order Sobol indices of parameter i, and $\text{interaction share}_i$ the part of i's influence that acts through other parameters.
- V_i is the variance contribution of parameter i and $V_{\sim i}$ the contribution of all the others.
- $Var\left[\log R_{\text{margin}}\right]$ is the total variance of the log margin.

$S_T \gg S_1$ marks a parameter that decides **how much everything else matters**.

Parameters

Nine parameters, each range carrying its own provenance; variance-based indices [22] with the total-order estimator of [23] as implemented in SALib [24]; Saltelli sampling, $n_{\text{base}} = 1024$, second order off, **11,264 total samples**, seed 20261008 (fixed, reproducible). Outputs: colonic mucosa, dysbiotic lumen, and the **bottleneck** =

$min_{\text{site}} R_{\text{margin}}$. The lactate ranges span both our published values and last cycle's generous ones, so the analysis covers the disagreement found in § 2.1.

Results

The prediction holds (Eq. (30)):

Output	$\sum S_1$	Residual interaction
Colonic mucosa	0.8405	0.1595
Dysbiotic lumen	**0.9173**	**0.0827**
Bottleneck	0.6248	**0.3752**

Single sites are nearly additive (8–16 % residual), exactly as the log transform should produce. The bottleneck's 37.5 % residual is genuine: taking a **minimum** is a real nonlinearity, not an artefact of multiplying. And n_Q's signature is as predicted — $S_T = 0.2314$ against $S_1 = 0.0688$ at the bottleneck, a ratio of 3.4.

> **This explains § 2.1 and § 2.2 at once: both inherited wide ranges are n_Q amplifying other uncertainties through a power law.** A single-index analysis cannot see that structure.

For the **colonic mucosa**, oxygen-sensor affinity dominates ($S_1 = 0.5873$, $S_T = 0.7365$), with the top three all oxygen parameters — independently confirming last cycle's finding. For the **bottleneck** the drivers are entirely different:

Parameter	S_1	S_T
repression strength	0.3317	**0.4568**
luminal EcN density	0.1386	**0.3548**
n_Q	0.0688	0.2314
oxygen-sensor affinity	0.0659	0.2055

> **Those two sets barely overlap** (the right panel above). The parameter that dominates the mucosa drops to 0.206 at the bottleneck, and the two that dominate the bottleneck do not appear in a mucosa-only analysis at all, since they enter only at luminal sites. **An analysis that stops at the mucosa is answering what drives a secondary site.**

A self-consistency check: repression strength has an interaction share of 0.004, i.e. it is a pure additive term in Eq. (28) — exactly what a multiplicative repression factor should be. A large interaction there would have indicated an implementation error.

Why this page is computed at the published sensor values

The Sobol analysis names K_{O_2} and n_Q as the two parameters that decide the margin. Both were carried over from last cycle as declared priors. Both turn out to have published measurements behind them, and **neither prior agreed with its measurement**, so on 2026-10-14 we replaced them and recomputed the page. This subsection is the record of that change.

Oxygen affinity. The closest characterisation is an FNR-family (ANR) promoter run in an oxystat across the full range, which reports half-maximal induction at about 5×10^2 Pa and states that the dose response closely resembles the *E. coli* FNR-activated promoters [10]. Against the same paper's air reading of 210×10^2 Pa, that is

$$K_{O_2} = \frac{5 \times 10^2}{101325} \times 100 = 0.49\,\% O_2$$

against the inherited 1.0 %, which was never sourced to a measurement.

Quorum Hill coefficient. A single-cell study of a LuxI/LuxR sender–receiver pair fits the pLux dose response at four levels of resolution and gets $n = 0.97 \pm 0.08$ in bulk, 0.95 ± 0.2 in a microfluidic chemostat, and 1.2 ± 0.3 to 1.5 ± 0.6 from single-cell trajectories; it also reports that earlier quantitative analyses of the AHL/pLux system "typically resulted in Hill exponents around $n = 1$–1.5" [11]. The inherited 2.0 was above that whole range. **We take the published centre, $n_Q = 1$, and sweep 0.95–1.5.**

What changed. Every headline number on this page moved, and all but one moved in our favour:

	Inherited priors	**Published values**
Bottleneck margin	19.64× mucosa	**21.85× mucosa**
Luminal sites, no native repression	0.46–1.21×	**0.23–0.60×**
Critical luminal density d_{crit}	6.80	**23.08**
Dose multiplier for a 150 µm lesion	8.23×	**2.82×**
Mucosa margin left after that escalation	1.36×	**4.41×**
150→1000 µm lesion turn-on	43,380×	195×

The bottleneck is larger, the lumen is *more* clearly off without the native repression, the density headroom triples, and the §3.5 collision — the worst result on this page — eases from "nothing left" to 4.4×. The one that shrinks is the lesion-size turn-on, which §2.2 already identified as an n_Q artefact rather than a mechanism; the **geometric floor it rests on, 178 µm, does not depend on either parameter and is unchanged.**

Across the published span the requirement is met with room. Sweeping n_Q over Ramalho's full range at Højberg's K_{O_2}:

	n_Q=0.95	n_Q=1.0	n_Q=1.05	n_Q=1.25	n_Q=1.5
Bottleneck	20.73×	**21.85**×	22.33×	19.56×	16.57×
Weakest link	mucosa	mucosa	lumen	lumen	lumen

The worst case is **16.57**×, 1.7× the requirement. What moves across the span is *which site* is the weakest link, not whether the design passes.

Where it does fail. Pushing either parameter well outside its published range breaks it: at K_{O_2}=2 % the mucosa collapses to ≈2.9× at every n_Q, and at n_Q=3 the dysbiotic lumen falls to 6.13×. Both are far outside anything reported, but they are the directions to watch, and they are why §7 still ranks a pPepT dose–response and an exogenous-AHL dose–response highly: a measurement **on our own parts** would replace a transferred one.

> **A note on transfer.** Neither measurement is on our construct. Ramalho's is a LuxI/LuxR pair in *E. coli*, which is the same regulator and promoter we use. Højberg's is an ANR promoter in *Pseudomonas*, which is a weaker transfer — same regulator family, different organism, and our promoter is pPepT specifically. We adopted it anyway because a measurement on a comparable part beats a declared value with no provenance, but it is the single assumption on this page we would most like to retire.

Where to spend effort. 81 % of the bottleneck variance sits on two parameters, and **one of them is measurable in a day** — the same 0 % vs 1 % O_2 comparison that §2.3.4 asks for. Luminal EcN density needs colonisation data we cannot generate this cycle; the closest that exists is the six-patient arm of [1], where two participants had no tumour enrichment over normal colon (§2.1). That is enough to say the parameter is real and not enough to narrow its range.

The lactate-prior dispute does not decide safety. All three lactate parameters sit below $S_T = 0.02$ at the bottleneck, because luminal lactate saturates the arm under either prior.

Coverage of the declared ranges:

Site	Protective fraction	median $\log_{10} R_{\text{margin}}$	5th pct
Tumour rim	**1.000**	1.385	0.654
Well-perfused tissue	**1.000**	2.243	1.345
Colonic mucosa	**0.992**	0.681	0.157
Colon lumen (healthy)	0.880	1.424	−0.593
Colon lumen (faecal)	0.777	0.861	−1.143
Colon lumen (dysbiotic)	0.744	0.728	−1.266
Bottleneck	**0.739**	0.361	−1.266

The safety argument holds across 73.9 % of the declared parameter space, and every failure in it is a luminal failure — independently reaching the same weak point as § 2.3's parameter-free result.

⚠ This is **not** a probability in any calibrated sense: the priors are ranges, not distributions. It is a coverage statement about declared ranges, and it rests entirely on the binary assumption of § 2.3.4 holding.

---

3. Efficacy — what dose is required?

3.1 Intra-nodule coverage and leakage

> **Question.** Of what is produced inside a nodule, how much is destroyed *outside* it? Split the degron's benefit into "makes less" and "spreads less". And is intra-nodule flatness a consequence of the diffusion lengths, as previously claimed?

Model

The fraction of production that is destroyed outside a lesion of radius R has a closed form, with g the exterior amplitude function of Eq. (37) and $\alpha = R/\lambda$:

$$E(R,\lambda)=\frac{3\,g(\alpha)(1+\alpha)e^{-\alpha}}{\alpha^{3}} \qquad (31)$$

$$G_{\text{field}}(R,\lambda)=\frac{\Psi_{\text{centre}}(R)}{\Psi_{\text{in}}(R;R)} \qquad (32)$$

$$\text{total off-target reduction}=\rho_{\text{pool}}\times\frac{E_{\text{untagged}}}{E_{\text{tagged}}}$$
$$\text{confinement share}=\frac{\ln(\text{confinement})}{\ln(\text{total})} \qquad (33)$$

where

- $E(R,\lambda)$ is the fraction of production destroyed outside a lesion of radius R.
- R is the lesion radius (µm) and λ the screening length of the effector field (µm).
- $\alpha=R/\lambda$ is the radius-to-screening-length ratio.
- $g(\alpha)$ is the exterior amplitude function of Eq. (37).
- $G_{\text{field}}(R,\lambda)$ is the core-to-rim ratio of the effector field inside the lesion.
- $\Psi_{\text{centre}}(R)$ is the normalised AHL at the lesion centre (Eq. (35)) and $\Psi_{\text{in}}(r;R)$ the normalised internal field (Eq. (36)).
- ρ_{pool} is the intracellular effector-pool ratio, untagged over tagged.
- E_{untagged} and E_{tagged} are the outside-destroyed fractions of the two constructs.

The share is taken in logs because the two contributions compose multiplicatively. **The closed form agrees with direct numerical integration to** 10^{-4}, which our tests check, so this section does not rest on hand algebra.

Results

At $R=400$ µm:

Species	λ	λ/R	**Destroyed outside**	**Free-field core:rim**	$\lambda<R$?
Lactate	5099 µm	12.75	**99.77 %**	1.539	✗
AHL	3000 µm	7.50	**99.36 %**	1.566	✗
GSDMD-N (no SsrA)	670.8 µm	1.677	**91.10 %**	1.784	✗
GSDMD-N (+SsrA)	**300 µm**	**0.750**	**73.16 %**	**2.072**	✅

Both constructs leak heavily. The 671 → 300 µm change sounds categorical — from larger than the nodule to smaller — but per molecule it moves off-target exposure by only 1.245×:

$$\text{retention} = \frac{1 - 0.7316}{1 - 0.9110} = 3.015\times, \text{confinement} = \frac{0.9110}{0.7316} = 1.245\times$$

$$\text{total} = 5.375 \times 1.245 = 6.693\times, \text{confinement share} = \frac{\ln 1.245}{\ln 6.693} = 11.5\,\%$$

> **About 88.5 % of the degron's spatial safety benefit comes from making less, and only 11.5 % from confining what is made.** The more quotable claim (retention improves 3.0×) is not the one that drives safety. This agrees with § 2.4, where the degron's value also lands on suppressing the accumulated pool.

Intra-nodule flatness is a boundary condition, not a length scale. Last cycle attributed core:rim ratios of 1.011–1.021 to the diffusion lengths dwarfing the nodule. That attribution does not hold: under free diffusion into surrounding tissue with the same decay rate, the same parameters give graded fields of 1.54–2.07×. **Even lactate, with $\lambda/R = 12.75$, shows a 1.54× gradient.** And the tagged effector's 300 µm range is *shorter* than the 400 µm nodule radius in any case, so its field should not be called a long-range average under any boundary condition.

What actually flattens the field is a **bounded domain with rim washout**. Both treatments are defensible — free field assumes equal decay inside and out, washout assumes a well-perfused rim carries effector away — and the truth sits between them, so **we report both and do not pick**. The therapeutic conclusion (flat coverage spares nothing, including the proliferating rim) survives; its *justification* changes from a scale argument to a boundary-condition argument.

3.2 Clustered micro-lesions

> **Question.** The AHL diffusion length is ~3000 µm, far larger than any single lesion. Does a cluster of micro-lesions 1–2 mm apart share one signal pool and switch itself on? If so, how large must the cluster be — and is that still micrometastasis?

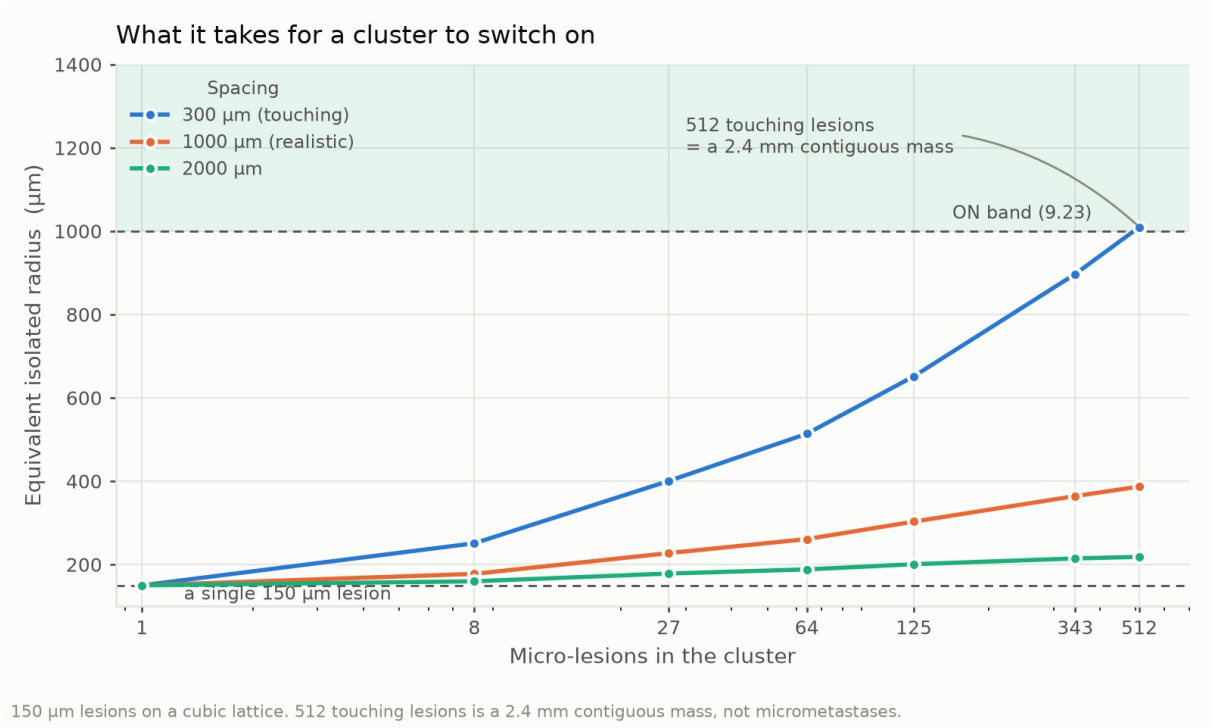


Figure 5. Equivalent isolated radius (Eq. (39)) of a cubic cluster of 150 µm lesions, against cluster size, at three spacings. **Answers § 3.2:** at realistic micrometastatic spacing no cluster leaves the OFF end of the band, and reaching the ON band takes 512 touching lesions — a contiguous millimetre-scale mass that the single-lesion model already says switches on. The multi-lesion extension opens no new risk window.

Model

The steady-state AHL equation is the screened (modified Helmholtz) form of the reaction–diffusion equation, whose spherically symmetric solutions are standard [25]:

$$D\,\nabla^2 C - \delta C + q(x) = 0,\ \lambda = \sqrt{D/\delta},\ \alpha = R/\lambda,\ s = r/\lambda \qquad (34)$$

Because **LuxI sits under pPepT rather than under pLux**, AHL production does not depend on AHL concentration. There is **no autoinduction loop**, so Eq. (34) is **linear**, superposition is **exact**, and an arbitrary three-dimensional arrangement of spherical sources is solved by summing closed-form single-sphere fields. We had planned a 2-D axisymmetric finite-difference solver; it is not needed, and the analytic result is more accurate, cheaper and easier to defend.

> **The same fact yields an architectural conclusion that was not previously stated: the quorum module is a *colonised-volume integrator*, not a bistable switch — no hysteresis, no sharp threshold, a graded response.** This agrees with § 2.2, where AHL integration carried 6.5× the weight of the hypoxia term.

Normalised to q/δ, for a single uniformly producing sphere:

$$\Psi_{\text{centre}}(R) = 1 - (1+\alpha)e^{-\alpha} \qquad (35)$$

$$\Psi_{\text{in}}(r) = 1 - (1+\alpha)e^{-\alpha}\frac{\sinh s}{s}, r \le R \qquad (36)$$

$$\Psi_{\text{out}}(r) = \lambda\, g(\alpha)\frac{e^{-r/\lambda}}{r}, g(\alpha) = \alpha e^{\alpha} - (1+\alpha)\sinh\alpha, r \ge R \qquad (37)$$

The two expressions match at $r = R$, and for small α the exterior field reduces to the screened (Yukawa) point-source kernel. **Our tests verify by finite difference that both branches satisfy the PDE.**

Cluster centre concentration and cooperation factor:

$$\Psi_{\text{cluster}}(i) = \Psi_{\text{centre}}(R) + \sum_{j \ne i} \Psi_{\text{out}}(d_{ij}), \text{coop} = \frac{\Psi_{\text{cluster}}(i)}{\Psi_{\text{centre}}(R)} \qquad (38)$$

Absolute AHL depends on β_Q / K_Q, on which last cycle's two calibrations differ 27-fold, so **we report no absolute concentration and issue no ON/OFF verdict.** Instead we invert Eq. (35):

$$R_{\text{eq}} : \Psi_{\text{centre}}(R_{\text{eq}}) = \Psi_{\text{cluster}} \qquad (39)$$

the **equivalent isolated radius** — the single lesion that would produce the same signal at its own centre. The threshold cancels, letting us compare directly against last cycle's 150 µm (OFF) → 1000 µm (ON) interval.

Finally, the small-α scaling that decides the dispersion question:

$$\left(\Psi_{\text{centre}} \overset{\text{own},\, \propto R^2}{\sim} \frac{\alpha^2}{2}\right) \text{vs} \left(\Psi_{\text{out}} \overset{\text{neighbour},\, \propto R^3/d}{\sim} \frac{R^3}{3\lambda^2 d}\right) \qquad (40)$$

where

- C is the AHL concentration, D its diffusion coefficient and δ its decay rate.
- $q(x)$ is the volumetric AHL production rate at position x.
- $\lambda = \sqrt{D/\delta}$ is the screening length.
- $\alpha = R/\lambda$ and $s = r/\lambda$ are the radius and the radial position in screening-length units.
- $\Psi_{\text{centre}}(R)$ is the normalised AHL at a single lesion's centre (Eq. (35)), $\Psi_{\text{in}}(r)$ the normalised internal field (Eq. (36)) and $\Psi_{\text{out}}(r)$ the normalised exterior field (Eq. (37)).
- $g(\alpha)$ is the exterior amplitude function of Eq. (37).
- $\Psi_{\text{cluster}}(i)$ is the normalised AHL at the centre of lesion i within a cluster.
- $\text{coop} = \Psi_{\text{cluster}}(i)/\Psi_{\text{centre}}(R)$ is the cooperation factor.
- d_{ij} is the centre-to-centre distance between lesions i and j.

- R_{eq} is the equivalent isolated radius: the single-lesion radius producing the same centre signal.

Substituting

$R = 150$ μm, $\alpha = 0.05$: $\Psi_{centre} = 1 - 1.05\,e^{-0.05} = 1.2091 \times 10^{-3}$. $R = 1000$ μm, $\alpha = 1/3$: 4.4625×10^{-2}.

> **Numerical note.** Eq. (35) and $g(\alpha)$ cancel to second and third order respectively, so below $\alpha < 0.01$ we switch to series expansions. The threshold must not be set higher: at $\alpha = 0.05$ the closed form still carries ~12 significant digits while a four-term series carries 5.

Results

Spacing	Lesions	Cooperation	**Equivalent radius**	Reaches ON?
300 μm (touching)	13	**4.74×**	**333.3 μm**	✗
	27	6.75×	400.6 μm	✗
	125	**16.91×**	**651.7 μm**	✗
1000 μm	13	1.89×	207.5 μm	✗
	125	3.95×	**303.0 μm**	✗
2000 μm	125	1.77×	200.8 μm	✗

No cluster reaches the 1000 μm equivalent. Cooperation is real but weak: at realistic micrometastatic spacing (⩾ 1 mm), 125 lesions reach only 201–303 μm.

How large would it have to be?

Touching cluster	Cooperation	Equivalent radius	Centre-to-centre span
$5^3 = 125$	16.91×	651.7 μm	1.2 mm
$7^3 = 343$	30.37×	897.0 μm	1.8 mm
$8^3 = 512$	**37.55×**	**1009.8 μm ← first ON**	**2.1 mm**

> **512 touching 150 μm lesions is a contiguous mass 2.1 mm across (2.4 mm including the outer radii). That is no longer micrometastasis — it is a tumour, and the single-lesion model already says a tumour of that size switches on. The multi-lesion extension opens no new risk window.**

That conclusion does not depend on the screening length

$\lambda = 3000$ µm is inherited from last cycle's run manifest and is the least defensible number in this section: it is $\sqrt{D/\delta}$ built from an AHL diffusion coefficient and an AHL decay rate, neither of which we could source to a direct measurement (§ 2.3 says the same about the decay half). Published tumour-tissue diffusion coefficients for a small metabolite are of order 10^{-5} cm^2 s^{-1} [26], which with an 8 h half-life puts λ in the low millimetres — consistent with 3000 µm, but as a plausibility check, not a citation for the value.

So the honest move is to show the conclusion over a range rather than defend the point estimate. Across a **ten-fold** span of λ:

λ	R_{eq} at 125 lesions	R_{eq} at 512	Lesions needed to reach 1000 µm
1000 µm	608.4 µm	888.7 µm	**1000**
3000 µm (ours)	**651.7 µm**	**1009.8 µm**	**512**
10000 µm	665.9 µm	1049.4 µm	**512**

No value of λ in that range lets a cluster reach the ON band at fewer than 512 lesions, and a shorter screening length makes it harder, not easier. The quantity the conclusion rests on is the *count*, and the count is bounded below across the whole span. That is why this section survives having its worst parameter undetermined, and it is the reason we report λ as scenario rather than quietly treating 3000 µm as known.

And dispersion is strictly worse than compaction, confirming Eq. (40):

Lesions	Equivalent radius (dispersed)	Volume-matched single	**Ratio**
4	200.4 µm	238.1 µm	**0.842**
8	251.0 µm	300.0 µm	**0.837**
27	400.6 µm	450.0 µm	0.890
125	651.7 µm	750.0 µm	0.869

> **Dispersing a given burden always produces less centre signal than compacting it (0.84–0.95×)**, because a lesion's own contribution scales as R^2 while a neighbour's scales as R^3/d.

Every modelling choice here is conservative in the same direction: steady state (transients fall below steady values), equal colonisation density across lesions (real micrometastases are probably colonised worse), no commensal lactonase degradation of AHL, and a lattice arrangement (which favours cooperation more than a random one).

For safety this strengthens the micro-lesion argument. For efficacy it says the circuit **cannot clear true micrometastases**, and distant disease has to be addressed immunologically rather than by colonisation — which § 3.6 takes up.

3.3 Dosing: schedule versus persistence

> **Question.** Can dosing schedule or inoculum size change cumulative exposure? If not, what is the actual lever?

Model

With a monomolecular approach to a fixed capacity (the shape § 1.3 established), cumulative exposure over a window T is

$$\int_0^T N\,dt = K\,T - (K - N_0)\frac{1 - e^{-r\,T}}{r} \qquad (41)$$

The inoculum enters only the second term, flattened by $1/r$. Against first-order clearance k_{clr} competing with growth μ:

$$K_{eff} = K\,max\left(1 - \frac{k_{clr}}{\mu}, 0\right), \text{exposure} = K_{eff}\,T \qquad (42)$$

where

- N is the bacterial density and T the exposure window (h).
- K is the carrying capacity and N_0 the inoculum density.
- r is the approach rate of the monomolecular law.
- k_{clr} is the first-order clearance rate and μ the growth rate.
- $K_{eff} = K\,max(1 - k_{clr}/\mu, 0)$ is the effective capacity under clearance.

Results

With $K = 1.507$, $r = 0.35$ h^{-1}, $T = 168$ h, a **32-fold range of inoculum (0.005 → 0.16) moves exposure by 0.178 %** — 248.885 to 249.327 — while onset time does shift, from 1.97 h to 1.66 h. **The dose changes onset, not the integral.**

The lever is persistence. At $\mu = 0.60$ h^{-1}:

Clearance	Plateau	Exposure	vs no clearance
0.00 h^{-1}	1.507	253.18	1.000
0.20 h^{-1}	1.005	168.78	0.667
0.55 h^{-1}	0.126	**21.10**	**0.083**
1.00 h^{-1}	0	0	eliminated

Clearance at 0.55 h^{-1} cuts exposure to 8.3 %, and above the growth rate the colony is eliminated. Clearance is the dominant lever; dosing frequency is not.

⚠ **But the corollary that redosing adds nothing is circular.** Our model contains **no mechanism by which a colony can go extinct**: growth is positive whenever the population is under capacity, there is no stochastic loss and no adaptive immunity. "Redosing merely reseeds an already-full niche" is therefore a property of how we built the model, not an inference from data. We have flagged it in code with three locked assertions (model_permits_extinction: False, conclusion_follows_from_construction: True, supported_by_data: False) so that it cannot be quoted as a result.

3.4 What dose is required

> **Question.** What dose makes the kill rate match proliferation — **in units we have actually realised**? Is it inside the measured range, and does the extrapolation stay within the guard rail we declared?

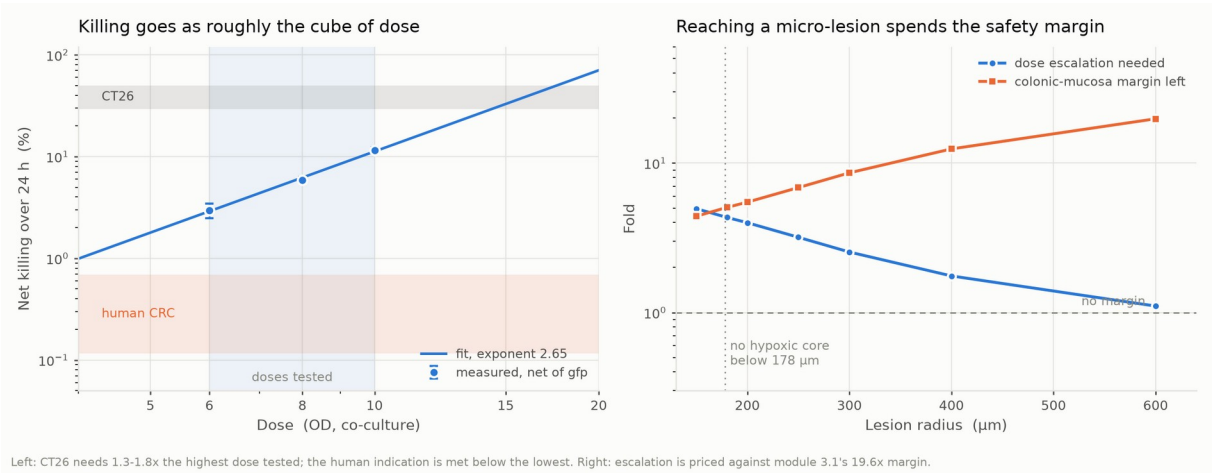


Figure 6. Left: net killing against dose with the fitted power law (Eq. (43)), against the requirement bands from Eq. (44). **The clinical indication is met below the lowest dose tested; CT26 needs 1.3–1.8× the highest.** Right: the two curves that collide in § 3.5 — the escalation a smaller lesion demands, and what survives of the 21.85× colonic-mucosa margin after paying for it. At 150 µm only 4.4× is left — below the 10× requirement, though no longer nothing.

Model

Step 1 — net dose–kill curve. Fitted in log–log after same-block, same-dose pairwise subtraction of a control:

$$f_{\text{net}}(d) = A\,d^{b} \;\Leftrightarrow\; \ln f_{\text{net}} = \ln A + b \ln d \qquad (43)$$

Step 2 — efficacy criterion. A lesion is held when kill rate matches proliferation rate $r = \ln 2 / T_{\text{double}}$. Over a 24 h exposure this is closed form and assumes nothing about dose–response shape:

$$f_{\mathrm{req}} = 100\left(1 - e^{-\ln 2 \cdot t_{\mathrm{exp}}/T_{\mathrm{double}}}\right)\% \qquad (44)$$

Step 3 — inversion.

$$d^{\dagger} = \left(\frac{f_{\mathrm{req}}}{A}\right)^{1/b} \qquad (45)$$

Delivery loss and lesion-radius compensation follow as cube roots:

$$M_{\eta} = \eta^{-1/b}, M(R) = \left(\frac{G(R)}{G(R_{\mathrm{ref}})}\right)^{-1/b} \qquad (46)$$

where

- $f_{\mathrm{net}}(d)$ is the net killing fraction at dose d (OD).
- A and b are the amplitude and exponent of the dose–kill power law (0.02510 %/OD^{b}, 2.651).
- f_{req} is the required kill fraction, set by the exposure window t_{exp} (24 h) against the tumour volume doubling time T_{double} (CT26 24–48 h; human CRC 100–600 d).
- $d^{\dagger} = (f_{\mathrm{req}}/A)^{1/b}$ is the required dose.
- η is the fraction of effector retained inside a nodule (0.268), and $M_{\eta} = \eta^{-1/b}$ the dose multiplier for effector loss outside it.
- $M(R) = \left[G(R)/G(R_{\mathrm{ref}})\right]^{-1/b}$ is the dose multiplier for the reduced gate output at radius R, with $G(R_{\mathrm{ref}})$ its value at the reference radius.

Parameters

Symbol	Value	Grade	Note
b	**2.651** (2.651–3.376)	**calibration**	interval endpoints = which control is subtracted
A	**0.02510** %/OD^{b}	**calibration**	log–log fit, $r^2 = 0.9956$
d	6 / 8 / 10 OD	**calibration**	triplicate
t_{exp}	24 h	protocol	endpoint readout
T_{double}	CT26 24–48 h; human CRC 100–600 d	scenario	brackets the measured median, 211 d (i.q.r. 112–404) on serial CT [27]
η_{nodule}	0.268	derived	§ 3.1: 73.2 % destroyed outside

Symbol	Value	Grade	Note
x_{max}	**3.0×**	**protocol, declared before use**	a power law has no threshold, so both directions are bounded

Substituting

The killing block carries two controls — the bare chassis and a plasmid-bearing strain expressing a non-effector protein — at all three doses. Subtracting the latter:

Dose	Raw	Control	**Net**	SE
6 OD	3.267 %	0.300 %	**2.967 %**	±0.494
8 OD	6.733 %	0.833 %	**5.900 %**	±0.368
10 OD	15.800 %	4.233 %	**11.567 %**	±0.855

Subtraction is not cosmetic: **at the top dose the chassis accounts for 26.8 % of the raw readout.**

Fitting Eq. (43) gives $b=2.651$, $A=0.02510$ ($r^2=0.9956$) net of the plasmid control, and $b=3.376$, $A=0.005049$ ($r^2=0.9992$) net of the bare chassis — **roughly a cube law**, with the interval coming from a structural choice rather than from fit error.

Required fractions from Eq. (44): CT26 at 24 h doubling gives $100\left(1-e^{-\ln 2}\right)=50.000\,\%$; at 48 h, 29.289 %; human CRC at 100 d (2400 h), $100\left(1-e^{-0.006931}\right)=0.6908\,\%$; at 600 d, 0.1155 %.

We report the clinical requirement at the **100 d** end throughout, which is the demanding one. The only series we could find that measured this directly — 43 colorectal cancers followed on serial CT without treatment — puts the median volume doubling time at **211 days**, interquartile 112–404 [27]. Our range brackets it, and the figure we quote is roughly twice as strict as that median, so the dose numbers below are not flattered by the choice.

Inverting (Eq. (45)), writing out the intermediate:

$$d^{\dot{c}}=\left(\frac{50.000}{0.02510}\right)^{1/2.651}=1992.1^{0.37725}=e^{0.37725\times 7.5969}=e^{2.8659}=17.57\,OD$$

Results

Target	f_{req}	**Required dose**	**Relative to highest tested**	Extrapolation	$\leqslant x_{max}$?
CT26, 24 h	50.000 %	15.26 – 17.57 OD	**1.53 – 1.76×**	1.53–1.76	✅
CT26, 48 h	29.289 %	13.02 – 14.36 OD	1.30 – 1.44×	1.30–1.44	✅
Human CRC, 100 d	**0.6908 %**	**3.49 – 4.29 OD**	**0.35 – 0.43×**	1.40–1.72	✅
Human CRC, 600 d	0.1155 %	1.78 – 2.53 OD	0.18 – 0.25×	2.37–**3.37**	⚠ one breach

⚠ The 600 d row, net of the plasmid control, requires extrapolating **downward** by 3.37×, outside the $x_{max}=3.0$ we declared before running anything. By our own rule it is not reportable. **But that row does not need the inversion at all** — see below.

The clinical indication is already met at every dose we ran, and that half of the answer needs no curve. Measured net killing at the lowest dose is 2.967 % against a requirement of 0.6908 %: **the lowest tested dose already exceeds the requirement by 4.3×.** This is a direct comparison, not an extrapolation.

Target	Verdict
Human CRC (both ends)	✅ **met at all three tested doses, no extrapolation needed**
CT26 (both ends)	❌ needs 1.30–1.76× escalation (inside the guard rail, reportable)

> **CT26 doubles two orders of magnitude faster than a human colorectal tumour and is by far the harder target. A shortfall in the mouse is not a shortfall in the indication.**

The exponent is what makes this usable on three points. Because dose enters near the third power, Eq. (45) takes a cube root of everything uncertain on the requirement side:

	Span
Required kill fraction across the four targets	**433.1×**
Corresponding required dose	**9.88×**
Compression	**43.8×**

A 600-fold range of doubling time becomes a 9.88-fold range of dose, and the 3.7-fold loss of effector outside a nodule costs only $0.268^{-1/2.651} = 1.64\times$ in dose. **The back-calculation is least sensitive to exactly the quantities we cannot measure** — a structural consequence of the exponent, not luck.

It does not produce an oral CFU figure. Two conversions are missing, neither in our workbook: a plate count of the co-culture, which would turn OD into bacteria per cancer cell; and the colonisation efficiency of an oral dose. The first is a **single plate** and would make every number here comparable to published colonisation densities — the highest-value remaining experiment in this project. The second needs animal work. The requirement is therefore reported in the unit we have actually realised, and our code asserts reports_oral_cfu = False so that a fabricated CFU number cannot drift onto this page.

3.5 Dose and safety are one constraint

Margins on the tissue axis are **ratios**, so raising the dose lifts both sides and leaves them unchanged. What a ratio does not track is the distance to an **absolute** tolerance. Smaller lesions give less gate output (Eq. (15)) and so need more dose (Eq. (46)), and that escalation is spent out of the 21.85× colonic-mucosa margin:

Lesion radius	$G(R)/G(400)$	Size multiplier	**Total escalation**	**Mucosa margin left**
400 µm	1	1.00×	**1.76×**	**12.44×** ✅
250 µm	0.20603	1.81×	3.19×	6.86× ⚠
180 µm	0.09156	2.46×	4.33×	5.05× ⚠
150 µm	0.06401	**2.82×**	**4.95×**	**4.41×** ❌

> **Treating a 400 µm nodule leaves an order of magnitude in hand. Reaching a 150 µm micro-lesion spends three quarters of the margin and drops it below the 10× requirement.**

This is the result that moved most when the sensor priors were replaced (§ 2.5). At the inherited $n_Q = 2$ the same escalation left **1.36×** — nothing at all. At the published exponent the gate output of a small lesion falls far less steeply with radius, so reaching one costs 2.82× rather than 8.23×, and 4.41× survives. **The conclusion is unchanged in kind and much less severe in degree:** a micro-lesion still cannot be reached without going under the requirement, so this is still a trade-off and not a free lunch.

The multipliers are large but finite — below the penetration depth the hypoxia arm falls to its basal level rather than to zero (Eq. (17)).

This meets § 3.2’ s cluster result from the opposite direction: quorum sensing cannot reach a micro-lesion, and dosing up until it does is not available either. Micrometastasis is not a dose problem. Both roads lead to § 3.6.

Converting the margin into a rate

A fold is not a safety statement. We convert it, using the **same machinery as the efficacy side** so that safety and efficacy sit on one ruler:

	Criterion	Rate form
Efficacy	kill rate ⩾ tumour proliferation	$\ln 2/T_{\text{double}}$ (doubling)
Safety	effector-driven epithelial loss ⩽ crypt renewal	$1/T_{\text{renew}}$ **(replacement)**

⚠ **The two rate forms differ** — $\ln 2/T$ for doubling, $1/T$ for replacement. Conflating them understates renewal by about 30 %.

$$f_{\text{tol}} = 100\left(1 - e^{-t_{\text{exp}}/T_{\text{renew}}}\right)\% = 100\left(1 - e^{-24/96}\right) = 22.12\,\%\ per\ 24\,h \qquad (47)$$

$$f_{\text{offtarget}} = f_{\text{target}}\, m^{-b}, m \geq \left(\frac{f_{\text{target}}}{f_{\text{tol}}}\right)^{1/b} \qquad (48)$$

where

- f_{tol} is the tolerable effector-driven epithelial loss fraction per 24 h, set by the exposure window t_{exp} (24 h) against the colonic epithelial renewal cycle T_{renew} (96 h — the midpoint of the 3–5 day range that human crypt kinetics support [28]; see below).
- $f_{\text{offtarget}}$ is the effector-driven epithelial loss fraction at an off-target site.
- f_{target} is the required kill fraction of the clinical indication (0.6908 %).
- m is the margin at the site, i.e. the relative gate output.
- b is the dose–kill exponent (2.651).

T_{renew} is a declared value, so it is worth checking whether anything depends on it. Human colonic crypts are about 82 cells high with a mid-crypt cycle time near 30 h [28], which is the measurement behind the 3–5 day whole-epithelium renewal that reviews quote; we take the midpoint. Across that whole range the constraint never comes close to binding:

T_{renew}	f_{tol} per 24 h	Margin required (Eq. (48))
3 d (72 h)	28.35 %	0.246×
4 d (96 h)	**22.12 %**	**0.270×**

T_{renew}	f_{tol} per 24 h	Margin required (Eq. (48))
5 d (120 h)	18.13 %	0.292×

Every one of those is below 1.0, i.e. renewal alone would tolerate the off-target dose even with *no* gate selectivity at all, and a 40 % swing in T_{renew} moves the requirement by 18 %. So the placeholder is not load-bearing, and this section's real content is the comparison below, not the threshold. We say so rather than presenting a derived number as though the margin had to clear it.

Substituting the clinical indication's demanding end ($f_{\text{target}} = 0.6908$ %):

$$f_{\text{offtarget}} = 0.6908 \times 21.852^{-2.651} = \frac{0.6908}{3556} = 1.94 \times 10^{-4}\,\%$$

Site	Margin	Implied epithelial loss / 24 h	Fraction of renewal
Colonic mucosa	21.85×	1.94×10^{-4} %	**1/113,800**
Colon lumen (dysbiotic)	23.08×	1.68×10^{-4} %	1/131,600
Tumour rim	48.66×	2.33×10^{-5} %	$1/9.5 \times 10^{5}$
Well-perfused tissue	101.7×	3.30×10^{-6} %	$1/6.7 \times 10^{6}$

Read with no amplification at all ($b = 1$, crediting the margin with nothing): $0.6908/21.85 = 0.0316$ % per 24 h, **still one part in 700 of renewal**.

Turned around, the loss criterion by itself asks for a margin of only **1.36**× (fitted exponent) or **2.26**× (linear, hardest target). **The axis delivers 21.85×, so the margin is not marginal against this criterion — and we do not need it to be.**

> ⚠ **But the criterion covers cell loss and nothing else.** GSDMD-N is a pyroptosis executioner, and IL-1β/IL-18 release is the entire basis of our distant-control argument, so a rate of pyroptosis far below renewal is **not automatically harmless** in colonic mucosa. Nothing here bounds that. **The loss criterion is therefore a lower bound on the margin we should want, not a statement that 21.85× is enough.** We report it because a fold with no criterion attached cannot be reviewed at all — not because it settles the question.
>
> The measurement that would bound it: **IL-1β release from a colonic epithelial line at the off-target gate output**, rather than at full induction.

3.6 The immune cascade

> **Question.** Under pMMR/MSS immunosuppression, how much pyroptotic flux does distant control require? Is there a region where **no** flux suffices? Is distant efficacy continuous in dose or threshold-like? And which half of the supervisor's local-versus-distant question can we actually answer?

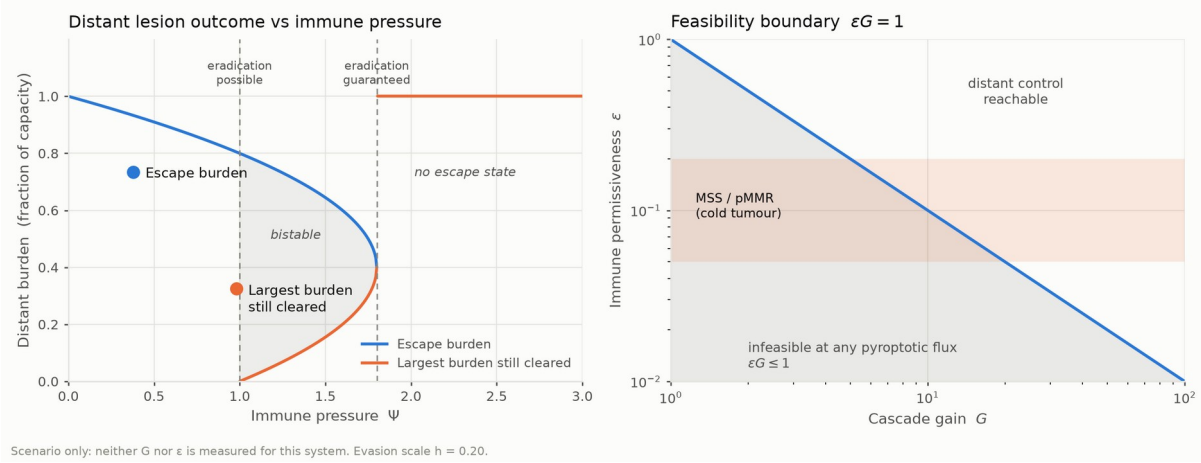


Figure 7. Left: equilibria of Eq. (51) against immune pressure Ψ, at h=0.2. Between Ψ=1 and Ψ_{crit}=1.80 the system is bistable, so there is a largest treatable distant burden — **this is why § 3.6 concludes that distant efficacy is threshold-like rather than continuous in dose.** Right: the εG=1 feasibility boundary (Eq. (52)); the shaded band is where MSS/pMMR sits, so the cascade gain must exceed roughly 5–20 before distant control is reachable at *any* flux. Scenario-grade throughout.

Since § 3.2 showed colonisation cannot reach disseminated disease and § 3.5 showed dose escalation cannot either, **this cascade is the only available route to distant control.** That promotes this section from optional to load-bearing. It is also the least constrained part of the project, so we keep it deliberately small and report **feasibility boundaries, not predicted efficacy**.

The premise the cascade rests on is not ours. Delivering active gasdermin into tumour cells with a bioorthogonal system, **pyroptosis of fewer than 15 % of tumour cells was enough to clear an entire 4T1 graft**, and the clearance was immune-dependent — it did not occur in immune-deficient mice or after T cell depletion [29]. That is the qualitative shape Eq. (49) to 6-7 encode: a sub-stoichiometric pyroptotic flux driving an immune response that does the rest, with a threshold rather than a dose-proportional effect.

What it does not give us is a number. That experiment is a different gasdermin delivery route, a different tumour model and a different species from ours, and it reports a sufficient fraction rather than a dose–response for antigen, DC maturation or priming. So we take from it only that the mechanism exists and is immune-mediated, and every rate constant below stays scenario-grade. § 7's

sixth item — a literature dose–response for IL-1β/IL-18 driving DC maturation — is what would move this section from feasibility toward numbers.

Model

$$\begin{aligned}\dot{A} &= \Phi - \delta_A A && \text{antigen + IL-1β/IL-18 signal}\\ \dot{D} &= a_D \frac{A}{K_A + A} - \delta_D D && \text{DC maturation, saturating}\\ \dot{T} &= a_T \varepsilon D - \delta_T T && \text{CTL priming, suppressed by } \varepsilon\\ \dot{M} &= r M\left(1 - \frac{M}{K}\right) - \frac{k T M}{h + M} && \text{distant lesion}\end{aligned} \qquad (49)$$

At steady state the upstream chain collapses. With $x = \Phi/(\delta_A K_A)$ the normalised pyroptotic flux, define the **immune pressure number** and the **cascade gain**:

$$\Psi = \varepsilon G \frac{x}{1+x}, G = \frac{k}{r h} \cdot \frac{a_T}{\delta_T} \cdot \frac{a_D}{\delta_D} \qquad (50)$$

Taking $K = 1$ and $r = 1$:

$$\dot{M} = M\left[(1 - M) - \frac{h\Psi}{h + M}\right] \qquad (51)$$

The entire cascade enters through one number. The only remaining parameter is h, the burden at half-maximal immune killing.

Because $x/(1+x) < 1$ always:

$$\varepsilon G > 1 \qquad (52)$$

This is an inequality, not an estimate. Above it, the required flux is

$$x_{min} = \frac{1}{\varepsilon G - 1} \qquad (53)$$

which diverges as $\varepsilon G \to 1^{+}$¿¿. Non-zero equilibria of Eq. (51) satisfy $M^2 - (1-h)M + h(\Psi - 1) = 0$ with $\Delta = (1-h)^2 - 4h(\Psi - 1)$:

$$M_{\pm} = \frac{(1-h) \pm \sqrt{\Delta}}{2}, \Psi_{\text{crit}} = 1 + \frac{(1-h)^2}{4h} \qquad (54)$$

where

- A is the antigen + IL-1β/IL-18 signal, Φ the pyroptotic flux driving it and δ_A its decay rate.
- D is the DC maturation signal; a_D, K_A and δ_D are the DC maturation rate, its half-saturation antigen level and the DC decay rate.
- T is the CTL population; a_T, ε and δ_T are the CTL priming rate, the priming suppression factor and the CTL decay rate.

- M is the distant-lesion burden; r and K its intrinsic growth rate and carrying capacity.
- k and h are the immune killing rate and the lesion burden at half-maximal killing.
- $x = \Phi / (\delta_A K_A)$ is the normalised pyroptotic flux.
- $G = \frac{k}{r\,h} \cdot \frac{a_T}{\delta_T} \cdot \frac{a_D}{\delta_D}$ is the cascade gain, and $\Psi = \varepsilon G x / (1 + x)$ the immune pressure number.
- $\Delta = (1-h)^2 - 4h(\Psi - 1)$ is the discriminant of the burden equilibrium, and $\Psi_{\text{crit}} = 1 + (1-h)^2/(4h)$ the critical immune pressure above which eradication is guaranteed.
- $M_{+¿¿}$ is the stable escape burden and M_{-} the basin boundary of the origin, i.e. the maximum treatable burden.

$M_{+¿¿}$ is the stable **escape** burden; M_{-} is the basin boundary of $M = 0$, i.e. the **maximum treatable burden**.

Results

Two thresholds, not one:

Threshold	Meaning
$\Psi > 1$	eradication **possible** — but only for a sufficiently small burden
$\Psi > \Psi_{\text{crit}}$	eradication **guaranteed**, from any starting burden

With $h = 0.2$, $\Psi_{\text{crit}} = 1 + 0.64/0.8 = 1.800$, and it is sharply h-dependent (5.51 at $h = 0.05$, 1.13 at $h = 0.5$). Between the thresholds the system is **bistable**:

Ψ	Escape burden $M_{+¿¿}$	Reduction	**Max treatable M_{-}**
1.0	0.800	0.200	0
1.5	0.645	0.355	**0.155**
1.800	**0.400**	**0.600**	**0.400** (roots meet, $\Delta = 0$)
2.0	0	1.000	1.000

The cold-tumour penalty. The indication is pMMR/MSS colorectal cancer, which our own design notes describe as poorly immunogenic, poorly infiltrated and immunosuppressive. **The one mechanism that must carry distant control is operating in its least favourable setting.** Required flux x_{min} (∞ = Eq. (52) fails):

$\varepsilon \setminus G$	2	5	10	30
1.00 (permissive)	1.000	0.250	0.111	0.0345
0.50	∞	0.667	0.250	0.0714
0.20	∞	∞	**1.000**	0.200
0.10	∞	∞	∞	0.500
0.05	∞	∞	∞	2.000

> **For ε in the range 0.05–0.2, the cascade gain must exceed roughly 5–20 before distant control is possible at all.** Converting a cold tumour to a hot one is the stated rationale for using pyroptosis; this is an estimate of how much conversion is needed. **And because our simplifications all err the same way — linear CTL priming with no clonal expansion, a constant ε, no immune memory — 5–20 is a lower bound.**

A small ε is the whole reason this section exists rather than being replaced by a checkpoint inhibitor. In the trial that established mismatch-repair status as the predictor, pembrolizumab gave an objective response in **0 of 18 mismatch-repair-proficient colorectal cancers**, against 40 % in the deficient arm [30]. Our indication is the 0-of-18 arm. We do not fit ε to that number — a response rate is not a priming suppression factor — but it is why we sweep ε at the suppressed end instead of a neutral one, and why the conclusion below is a feasibility boundary rather than a predicted benefit.

Clinical implication of bistability: there is a maximum treatable distant burden, so **distant efficacy is not a continuous function of dose**. Whether the burden is inside the basin when treatment starts matters more than how much is given. **Treating early beats treating more** — a stronger statement than §3.3's "clearance matters more than frequency".

Answering the local-versus-distant question honestly — one half is answerable and one is not:

	Status
Local	✅ **measurable**: 15.8 % killing at 10 OD for the construct we build
Distant	❌ **not measurable; this model gives feasibility only, never a number. Any specific percentage would be invented.**

So the honest answer is three sentences: distant killing is the **only** mechanism that can address disseminated disease; in MSS/pMMR it requires a cascade gain above 5–20, which is not a low bar and is a lower bound; and it is **threshold-like**, with a maximum treatable burden, so early treatment beats more treatment.

⚠ **This whole section is scenario-grade.** Neither G nor ε nor h nor x is measured, and nothing in our dataset could check them. The cheapest step toward numbers is **literature work, not wet work**: anchoring x to an IL-1β/IL-18 → DC maturation dose response. Until that exists, this section reports no absolute efficacy figure.

---

4. Validation

> **Question.** The evidence grade validation has been empty, on the grounds that there is no second dataset. Does that reason hold? Which conclusions can actually be put at risk — and does the protocol have the power to reject anything?

A held-out prediction does not need a second dataset. It needs a fit denied part of the first one, then asked to produce the part it was denied. Four such tests exist in our data, each one bearing on a conclusion the project actually uses, and **all four pass**. Tolerances were fixed as module constants before any test was run, and a fifth entry is included precisely because it fails, to show the protocol can reject. This is internal validation and we call it that: there is one dataset and there will not be another this cycle.

$$\left|\frac{\hat{y}}{y}-1\right|\le \mathrm{tol}_{\mathrm{ratio}}=0.25\,,¿\,\hat{y}-y\vee\frac{¿}{SE}\le \mathrm{tol}_{\mathrm{SE}}=2 \qquad (55)¿$$

Test	n	Passed	Worst ratio	**What a failure would have falsified**
Dose–kill curve, leave one dose out	6	**6**	1.165	the back-calculated dose (§ 3.4)
Gate corners at 24 h, leave one out	4	**4**	1.022	**the AND-logic claim (§ 1.1)**
Gate corners at 9 h (**expected to fail**)	4	**0**	4.447	nothing — this is the power check

Test	n	Passed	Worst ratio	**What a failure would have falsified**
24 h capacity forecast from $t \le 12$ h	8	**8**	1.080	every density on the tissue axis (§ 2.1, § 2.4)
Dose–kill exponent refitted per construct	6	**6**	—	the cube law, hence § 3.4's robustness argument
Total	**28**	**24 pass, 4 reject by design**		

4.1 The strongest test: the gate predicts its own corners

A multiplicative gate means the four corners are additive in probit units, so **any three determine the fourth**:

$$\hat{z}_H = z_i + z_j - z_k, \hat{p}_H = \Phi(\hat{z}_H) \qquad (56)$$

where

- z is the probit-transformed positive fraction and p the positive fraction of cells above the gate.
- $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution.
- $\hat{z}_H$ and $\hat{p}_H$ are the probit value and positive fraction predicted for the held-out corner H.
- z_i, z_j and z_k are the probit values of the three observed corners.

Holding out the fully-on corner and predicting it from the other three:

$$\hat{z}_D = 0.9525 + (-0.4761) - (-1.6044) = 2.0808 \Rightarrow \hat{p}_D = \Phi(2.0808) = 98.13\,\%$$

against an observed **98.08 %** — an error of 0.05 percentage points.

All four corners pass, worst ratio 1.022. The AND claim is therefore an out-of-sample prediction, not only an internal consistency test: the held-out corner was not used to produce the number that predicts it.

The test is run at the mid-course timepoints too, and there it is designed to fail. § 1.1 established that the gate is super-multiplicative while it is still turning on, and concluded that the AND claim belongs to the endpoint. That is a statement

about where this model applies, made before the holdout was run. The mid-course rows check it:

Timepoint	Corners passing	Worst ratio	Status
6 h	0 / 4	1.60	outside the claimed domain
9 h	0 / 4	4.45	outside the claimed domain
24 h	**4 / 4**	**1.022**	**the claim**

> **The boundary falls exactly where § 1.1 put it**, which is what makes the 24 h agreement worth something — § 4.2 takes that argument up.

Everything downstream consumes the 24 h factorisation — § 2.1's tissue axis, § 2.2's size turn-on, § 3.4's dose. and_logic_holds_at_the_endpoint_only() asserts all three legs (the endpoint predicts, the mid-course does not, the holdout has power), so the scope of the claim cannot quietly decay.

4.2 The protocol has power, because it rejects

The 9 h holdout **fails on all four corners at 3.24 standard errors, and it is meant to.** The 6 h holdout fails too.

> **A protocol that passed everywhere would have no power, and its passes would carry no information.** The 9 h timepoint carries a genuine interaction term (§ 1.1: $+1.135 \pm 0.350$, $p \approx 0.01$), so the holdout is *obliged* to reject it. We assert "endpoint passes **and** mid-course fails" as a single test.

A methodological by-product: the 6 h corners **pass** the SE criterion ($\pm 1.18 < 2$) but **fail** the ratio criterion ($0.488\times$). **The two criteria in Eq. (55) are not redundant** — the SE criterion is lenient where replicate scatter is large, and the ratio criterion is not. Both are needed.

4.3 The capacity forecast, and a bias that cancels

Forecasting the 24 h plateau from $t \le 12$ h **overshoots by 5.0 % across all eight curves**, consistent with the creep phase that no three-parameter sigmoid reproduces (§ 1.3). All eight still pass.

The interesting part is what happens to the quantity the degron decision actually uses. Eq. (23) rests on a **ratio**, not on either plateau:

$$\hat{r}_{\text{biomass}} = \frac{\beta_{\text{untagged}} K_{\text{untagged}}}{\beta_{\text{tagged}} K_{\text{tagged}}} \to r_{\text{biomass}} \text{ when } \beta_{\text{untagged}} \approx \beta_{\text{tagged}} \qquad (57)$$

Condition	Forecast ratio	Observed	Error
Aerobic + lactate (the one the criterion uses)	0.58712	**0.58475**	**0.41 %**
Anaerobic + lactate	0.88809	0.91593	3.04 %

> **Both strains are read high by nearly the same factor, so the ratio is reproduced far better than either of its components — and the ratio is what the decision depends on.** This is out-of-sample evidence for the design choice of working in ratios rather than absolute values.

4.4 The exponent, refit per construct

Construct	b	r^2	Carries effector?	In headline fit?
gfp	**5.105**	0.9591	❌	❌
GSDMD	**3.956**	0.9806	✅	❌
gfp-TAT	3.066	0.9448	❌	❌
GSDMD-SsrA	**3.059**	**0.9856**	✅	✅
GSDMD-TAT	2.329	0.9994	✅	❌
EcN	**2.230**	0.7627	❌	❌

Five of these six are data the headline fit never used. **All six exceed 2**, and the headline interval 2.651–3.376 sits inside the spread. Point values are not transferable (they span 2.3×), but the **sign of the curvature** is:

> **Even the constructs carrying no effector at all are supra-linear, so "dose cubed" is a property of how dose enters this system, not of the effector.** That is why §3.4's answer is insensitive to which control is subtracted.

4.5 What this validation does and does not cover

- **Not external validation.** One workbook; all holdouts are internal.
- **It cannot detect an error shared by the whole dataset** — a calibration offset, or a block-level scaling factor. **We know that class of error is real here, because we found one** (§5.1), and we found it by checking internal consistency. Cross validation would never have shown it.

- **It touches no scenario parameter.** n_Q, λ_{AHL}, the repression strengths, G, ε — none of these are in the data, so none can be held out.
- **What it does rule out** is a fit that only reproduces the points it was handed, which is the failure mode three-point curves and four-corner designs are most prone to.

The validation column in our evidence table therefore reads **"internal cross validation, one workbook"** — not "independent dataset".

---

5. Limitations: what this model cannot tell you

We have written the assumptions and gaps out module by module in the repository. This section is the summary. Where a limitation changes a specific conclusion we have also said so next to that conclusion. Every limitation here that an experiment could settle is costed and ranked in § 7.

5.1 The data we have is all the data there is

Our workbook was frozen on 8 October. We asked the wet lab whether anything else was coming and the answer was no: **no animal work this cycle, no lactate gradient, no inactive-GSDMD control, no empty-vector control, and nothing at 0 % oxygen.** The gaps below are permanent for this year, not a to-do list.

Three specific holes in the blocks we do have.

The reporter intensity block records undetected measurements as 0, and there are 19 of them. Sixteen are contradicted by the percent-positive block measured on the same samples, which reports 2–32 % positive cells where the intensity read zero. Those zeros are the instrument's floor, not darkness. We keep them in the means and we refuse to let them be denominators. One group — 1 % oxygen with lactate at 24 h — reads 40433, 32134 and 0; we print all three, because a mean with an SD would imply a noise model that is not there.

Two measurements are missing. GSDMD at 8 OD has $n=2$, and GSDMD-TAT against L929 has $n=2$ with both values exactly 10.0, so its SD is zero and no t test exists for that pair. Separately, GSDMD at 10 OD reads 14.3, 26.0 and 26.7, where the low measurement is 55 % of the other two. That group is the control for our SsrA argument, so we have used it as a range.

We also found a block-level error, and it is worth stating plainly. Blocks 5 and 6 sit about 7.6-fold apart for every construct measured independently in both, and two of the block-6 cancer rows are block-5 numbers copied across, replicate for replicate. Every cross-block ratio built on those rows is withdrawn — including a

selectivity index of 11.4× and a potency cost of 4.70× that would have **inverted** the SsrA decision (§ 2.4). One convention now forbids cross-block pairing, and it retired both numbers at once.

5.2 Nothing in our data identifies a sensor parameter

The reporter experiment has two oxygen levels and two lactate levels. Two points do not locate a midpoint or a slope. **Every EC_{50}, Hill coefficient and basal level in this model is therefore a literature prior or a declared value**, including the three the gut-lumen ceiling depends on. They are all marked scenario in § 6.

The one we would most like to have is **the quorum Hill coefficient**. It multiplies everything in the hypoxia and density path, its total-order Sobol index at the bottleneck is 3.4× its first-order index, and **both of the wide ranges we inherited from last cycle turn out to be that one coefficient rather than a measurement spread** (§ 2.1, § 2.2). An exogenous AHL dose response on pLux would fix it in an afternoon. We did not get to it.

Two of these priors are not as unconstrained as that paragraph implies, and we would rather say so than let the reader assume the worst. n_Q and K_{O_2} — the two the Sobol analysis puts first — both have published measurements on comparable parts, and § 2.5 evaluates the model at them. The honest statement is narrower than "nothing identifies a sensor parameter": **nothing in *our* data identifies one, and for two of them a published measurement on a comparable part is better than the value we inherited — so we adopted it.** n_Q and K_{O_2} are graded literature_prior in § 6, and every number on this page is computed at them. § 2.5 records what the inherited priors gave, so the change is auditable rather than silent.

One more of this kind. The fold changes from the probit inversion — 17× for oxygen and 642× for lactate — are proportional to the cell-to-cell spread of expression, and a group mean plus a positive fraction cannot resolve that spread. We report them as conditional on it and build nothing on them. **The multiplicativity test itself does not use it, which is why that is the claim we lead with.**

5.3 The assumption that would change the answer

This has its own section above (§ 2.3.4). The short version: the luminal safety argument turns on the colon lumen being anoxic while the tumour core is not, and we have data at 1 % oxygen and none at 0 %. If the native repression does not engage at 0 %, the luminal sites go from 23–60× to 0.23–0.60× and the bottleneck moves, along with the recommendation against a de-repressed promoter.

Every other open parameter here widens an interval when it is wrong. This one flips a sign. One plate settles it.

5.4 Where the modelling is thin

The dose–kill curve is three points. It fits well (r^2>0.995), but three points say nothing about a plateau or a threshold, so the inversion refuses to report anything more than threefold outside the measured range, in either direction — and § 3.4 flags the one row that breaches it.

Killing in culture over 24 h is not an in vivo kill rate. Every cell in the culture meets the bacteria; in a tumour only the colonised part does. We handle this with a declared delivery efficiency and show the cube law makes the answer insensitive to it — but "insensitive to a number we made up" is still a number we made up.

The off-target margin is converted using epithelial renewal, which covers cell loss and nothing else. GSDMD-N is a pyroptosis executioner and the IL-1β/IL-18 release it causes is the entire basis of our distant-control argument, so a rate of pyroptosis well under the renewal rate is not obviously harmless in colonic mucosa. **We have no bound on that at all.**

The immune cascade is scenario-grade from start to finish. None of its parameters are measured and nothing in our dataset could check them. It gives feasibility boundaries and we have tried hard not to let it read as a prediction.

Our dosing model contains no mechanism by which a colony can go extinct, so the conclusion that redosing adds nothing is a property of how we built the model (§ 3.3). It is flagged in code so it cannot be quoted as a result.

Two boundary conditions, no decision between them. Intra-nodule flatness differs 1.5–2× between the free-field and washout treatments (§ 3.1). We report both, which means "how flat is the field inside a nodule" is a question this project answers only as an interval.

Oxygen is in gas-phase percent throughout, because that is how the reporter was calibrated, at 21 % and 1 %. Tissue pO_2 is lower. The directional statements survive the conversion; any absolute number would need it done properly.

5.5 Things we are still unhappy about

The selectivity index is 3.3×, from one experiment at n=3. It is the only protein-level selectivity figure that survived our audit and it is not a large number. The safety case rests on the circuit's tissue selectivity — which is what an AND gate is for — but we would be happier if the effector pulled more weight.

We cannot explain why the chassis-only controls curve up with dose nearly as steeply as the effector constructs. EcN alone kills 1.1, 1.2 and 3.6 % across the three doses, and the GFP strain 0.3, 0.8 and 4.2 %. Something about dense bacteria is killing CT26 in this assay and it is not the effector. Subtracting it keeps the net analysis sound, and § 4.4's per-construct refit turns the puzzle into a result — supra-linearity is a property of the assay, not the effector — but we still do not know the mechanism.

Last, the tidiness bothers us a little. The cluster result and the dose result reach the same boundary from opposite directions. The growth forecast's bias cancels in exactly the ratio the degron decision depends on. The Sobol residuals land where the log-scale argument predicts. We have checked all of these and they hold. But **a model in which everything agrees is a model that ought to be checked by someone who did not build it**, and that has not happened yet.

---

6. Parameters

Every number the model uses, with its unit, its bounds where it has them, how it was arrived at and what it is for.

The table is generated from config/parameters.csv, and src/parameters.py resolves each row against the constant the code actually uses and fails the test suite if the two disagree, so neither this page nor the published one can hold a stale copy.

Five grades rather than the usual three. **Calibration** is fitted to our data, **literature prior** is measured by someone else on a comparable system, **scenario** is a value we declared, **derived** is computed from other entries, and **protocol** is a threshold we fixed before using it.

The literature-prior grade was added on 2026-10-14, when two parameters the sensitivity analysis puts first — n_Q and K_{O_2} — turned out to have published measurements behind them and were promoted out of scenario (§ 2.5). Keeping the two apart matters: "somebody measured this on a similar part" and "we picked this" are different claims, and the old four-grade scheme printed them identically.

The census is worth stating plainly: 20 of 36 entries are scenario-grade, 2 carry a published measurement, and only 5 are fitted to our own data. That is the honest shape of a one-cycle model, and it is why the load-bearing arguments here are built to hold **for any value of the parameters they involve**:

Load-bearing result	Holds for any…	Section
Structural anti-selectivity in the lumen (Eq. (18))	EC_{50}, Hill, basal	§ 2.3
Analytic ceiling on lactate discrimination (Eq. (19))	EC_{50}	§ 2.3
Geometric micro-lesion floor (Eq. (17))	sensor parameter	§ 2.2
Exact superposition of AHL fields (Eq. (34))	source arrangement	§ 3.2
SsrA keep/drop criterion (Eq. (23))	dose–response shape	§ 2.4
Multiplicativity test (Eq. (3))	σ	§ 1.1
Immune feasibility ceiling (Eq. (52))	flux x	§ 3.6

The five calibrated entries are the ones every design decision rests on.

Symbol	Parameter	Value	Unit	Evidence	Source
EC50_lac	lldPRD/ LldR half-maximal lactate	1.6 [0.5, 5]	mM	scenario	Zuniga et al. 2021 (ALPaGA) [8]: wild-type lldPRD biosensor, EC50 ~1.6 mM
n_lac	lldPRD Hill coefficient	1 [1, 2]	dimensionless	scenario	No cooperativity reported for LldR; single-site Hill is the default. 9.23 assumed 2
b_lac	lldPRD basal fraction	0.142857 [0.001, 0.2]	fraction of max	scenario	~7-fold dynamic range

Symbol	Parameter	Value	Unit	Evidence	Source
					implies basal ~1/7; 9.23 used 0.0078
K_O2	pPepT/FNR half-maximal oxygen	0.49 [0.3, 3]	% O2 (gas phase)	literature_prior	Hojberg et al. 1999 [10]: FNR-family promoter half-maximal at 5e2 Pa = 0.49 % O2. 9.23 used 1.0 %, unsourced. Dominant driver at colonic mucosa (ST 0.736) – see 2.5
n_O2	pPepT Hill coefficient	2 [1, 3]	dimensionless	scenario	Shared prior with 9.23
b_O2	pPepT basal fraction	0.05 [0.02, 0.2]	fraction of max	scenario	9.23 fitted 0.122 (its pept_basal_fraction)
n_Q	pLux quorum Hill coefficient	1 [0.95, 1.5]	dimensionless	literature_prior	Ramalho et al. 2016 [11]: pLux dose response fitted at four resolutions, n = 0.95-1.5, centre

Symbol	Parameter	Value	Unit	Evidence	Source
					1.0. 9.23 used 1.0 and 2.978 – see 2.5
L_AHL	AHL screening length	3000	um	scenario	9.23 run_manifest.json
d_O2	oxygen penetration from perfused surface	178	um	derived	9.23 MODEL_SPEC_4_3 section 2: 1 % O2 contour at 222 um in a 400 um nodule
R_nodule	reference nodule radius	400	um	scenario	9.23 used this throughout
O2_rim	oxygen at the perfused rim	5	% O2	scenario	9.23 spatial parameters
O2_core	oxygen in the hypoxic core	0.5	% O2	scenario	9.23 spatial parameters
R_off	lesion radius 9.23 reported OFF	150	um	scenario	9.23 finding 4
R_on	lesion radius 9.23 reported ON	1000	um	scenario	9.23 finding 4
L_lac	lactate diffusion length	5099	um	scenario	9.23 run_manifest.json
L_eff_untag	GSDMD-N	670.8	um	scenario	9.23

Symbol	Parameter	Value	Unit	Evidence	Source
ged	diffusion length without SsrA				run_manifest.json
L_eff_tagged	GSDMD-N diffusion length with SsrA	300	um	scenario	9.23 run_manifest.json
delta_nat	effector decay without the degron	0.08	1/h	scenario	Half-life 8.7 h in normal tissue
delta_built	effector decay with the degron as built	0.35	1/h	scenario	Half-life 1.6 h; ~2.3x stronger than the optimum
rho_pool	intracellular pool ratio untagged over tagged	5.375	dimensionless	derived	Module 3.4 TradeoffModel
eta_nodule	tagged effector retained inside a 400 um nodule	0.268	fraction	derived	Module 4.3: 73.2 % destroyed outside
b_dose	net dose-kill exponent (net of gfp)	2.65078 [2.65078, 3.3764]	dimensionless	calibration	Block 5, GSDMD-SsrA net of gfp, log-log fit, r2 0.9956. Bounds are the two control choices
A_dose	net dose-kill amplitude	0.0250985	% per OD^b	calibration	Block 5, same fit

Symbol	Parameter	Value	Unit	Evidence	Source
	(net of gfp)				
r_biomass	induced 24 h OD untagged over tagged	0.584746	dimensionless	calibration	Blocks 1-2, aerobic + lactate. Forecast from t <= 12 h reproduces it to 0.41 %
r_potency	per-OD killing untagged over tagged	1.4135	dimensionless	calibration	Block 5 at 10 OD, both constructs in the same block
SI	selectivity index CT26 over L929	3.32558	dimensionless	calibration	Block 6 internal pairing only, n=3, p < 0.001. The 11.4x and 2.97x figures are withdrawn
T_CT26	CT26 volume doubling time	[24, 48]	h	scenario	Murine line; two orders of magnitude faster than the indication
T_human	human colorectal volume doubling time	[100, 600]	d	scenario	The clinical indication
T_renew	colonic epithelial renewal cycle	[3, 5]	d	scenario	Crypt-to-surface replacement. A cycle,

Symbol	Parameter	Value	Unit	Evidence	Source
					so the rate is 1/T and not ln2/T
h	burden at half-maximal immune killing	0.2	fraction of capacity	scenario	Declared; module 6 is scenario_only throughout
t_exp	co-culture exposure window	24	h	protocol	Block 5 is an endpoint readout
x_max	maximum reportable extrapolation	3	fold	protocol	Declared before use; a power law has no threshold, so both directions are bounded
b_cons	conservative exponent for margin claims	1	dimensionless	protocol	Credits the margin with no amplification at all
tol_ratio	held-out prediction ratio tolerance	0.25	fraction	protocol	Declared before the validation ran
tol_se	held-out prediction standard-error tolerance	2	SE	protocol	Declared before the validation ran
t_cut	forecast cutoff for the capacity	12	h	protocol	Must exclude the 24 h target

Symbol	Parameter	Value	Unit	Evidence	Source
	holdout				

7. Future improvements: the experiments we would run next

Ranked by information per unit cost, from our own sensitivity analysis rather than from intuition.

#	Experiment	Cost	What it settles
1	**Reporter at 0 % vs 1 % O_2, lldPRD-only**	**one plate**	§ 2.3.4's sign-change assumption — 46 % of bottleneck variance (§ 2.5)
2	**Plate count of the co-culture** (6/8/10 OD)	**one plate**	converts every dose in § 3.4 from "relative to 10 OD" into MOI, comparable to published colonisation densities
3	**Exogenous AHL dose response on pLux**	one afternoon	n_Q — the single parameter behind both inherited wide ranges (§ 2.1, § 2.2)
4	**Per-sample spread of log-fluorescence** (CV or robust SD, gated and ungated)	one cytometry run	identifies σ, turning § 1.1's σ-conditional folds into measurements
5	IL-1β release from a colonic epithelial line at off-target gate	one plate + ELISA	the only bound on § 3.5's inflammation gap

#	Experiment	Cost	What it settles
	output		
6	Literature: IL-1β/IL-18 → DC maturation dose response	desk work	anchors x in § 3.6, moving the cascade from feasibility toward numbers

Four of the six are a single plate or an afternoon, and each one is here because the sensitivity analysis named it, not because it seemed interesting. That ranking is itself a model output: § 2.5 shows 81 % of the bottleneck variance sitting on two parameters, and items 1 and 3 are those two parameters.

---

8. References

Numbered in order of first appearance. Every entry was opened and checked against the original article; config/references.csv records the date, and the numbers on this page are generated from that file at build time, so a citation cannot drift out of step with its marker.

1. Gurbatri CR, Radford GA, Vrbanac L, Im J, Thomas EM, Coker C, et al.. Engineering tumor-colonizing E. coli Nissle 1917 for detection and treatment of colorectal neoplasia. *Nature Communications* 2024; 15:646. doi:10.1038/s41467-024-44776-4. PMID 38245513
2. Stritzker J, Weibel S, Hill PJ, Oelschlaeger TA, Goebel W, Szalay AA. Tumor-specific colonization, tissue distribution, and gene induction by probiotic Escherichia coli Nissle 1917 in live mice. *International Journal of Medical Microbiology* 2007; 297(3):151–162. doi:10.1016/j.ijmm.2007.01.008. PMID 17448724
3. Lim R. BBa_M36021: lldPRD operon promoter. *iGEM Registry of Standard Biological Parts* 2014; Stanford BIOE44-S11; 332 bp basic part, derived from BBa_K822000. registry.igem.org/parts/bba-m36021
4. Finney DJ. Probit analysis. *Cambridge University Press* 1971; 3rd edition, 333 pp.
5. Zwietering MH, Jongenburger I, Rombouts FM, van ' t Riet K. Modeling of the bacterial growth curve. *Applied and Environmental Microbiology* 1990; 56(6):1875–1881. doi:10.1128/aem.56.6.1875-1881.1990
6. American Type Culture Collection. CT26.WT (ATCC CRL-2638): mouse colon carcinoma, BALB/c. *ATCC* 2026; atcc.org/products/crl-2638

7. American Type Culture Collection. NCTC clone 929 [L-929] (ATCC CCL-1): mouse subcutaneous connective tissue fibroblast, C3H/An. *ATCC* 2026; atcc.org/products/ccl-1
8. Zúñiga A, Camacho M, Chang H-J, Fristot E, Mayonove P, Hani E-H, Bonnet J. An engineered lactate-responding promoter system operating in glucose-rich and anaerobic environments. *ACS Synthetic Biology* 2021; 10(12):3527–3536. doi:10.1021/acssynbio.1c00456
9. Goers L, Ainsworth C, Goey CH, Kontoravdi C, Freemont PS, Polizzi KM. Whole-cell Escherichia coli lactate biosensor for monitoring mammalian cell cultures during biopharmaceutical production. *Biotechnology and Bioengineering* 2017; 114(6):1290–1300. doi:10.1002/bit.26254. PMID 28112405
10. Højberg O, Schnider U, Winteler HV, Sørensen J, Haas D. Oxygen-sensing reporter strain of Pseudomonas fluorescens for monitoring the distribution of low-oxygen habitats in soil. *Applied and Environmental Microbiology* 1999; 65(9):4085–4093. doi:10.1128/AEM.65.9.4085-4093.1999. PMID 10473420
11. Ramalho T, Meyer A, Mückl A, Kapsner K, Gerland U, Simmel FC. Single cell analysis of a bacterial sender-receiver system. *PLoS ONE* 2016; 11(1):e0145829. doi:10.1371/journal.pone.0145829
12. Thomlinson RH, Gray LH. The histological structure of some human lung cancers and the possible implications for radiotherapy. *British Journal of Cancer* 1955; 9(4):539–549. doi:10.1038/bjc.1955.55. PMID 13304213
13. Worth KR, Papandreou I, Hammond EM. How the histological structure of some lung cancers shaped almost 70 years of radiobiology. *British Journal of Cancer* 2023; 128(3):407–412. doi:10.1038/s41416-022-02041-9. PMID 36344595
14. Walenta S, Schroeder T, Mueller-Klieser W. Lactate in solid malignant tumors: potential basis of a metabolic classification in clinical oncology. *Current Medicinal Chemistry* 2004; 11(16):2195–2204. doi:10.2174/0929867043364711. PMID 15279558
15. Jørgensen VL, Reiter N, Perner A. Luminal concentrations of L- and D-lactate in the rectum may relate to severity of disease and outcome in septic patients. *Critical Care* 2006; 10(6):R163. doi:10.1186/cc5102. PMID 17116255
16. Duncan SH, Louis P, Flint HJ. Lactate-utilizing bacteria, isolated from human feces, that produce butyrate as a major fermentation product. *Applied and Environmental Microbiology* 2004; 70(10):5810–5817. doi:10.1128/AEM.70.10.5810-5817.2004. PMID 15466518
17. Mayeur C, Gratadoux J-J, Bridonneau C, Chegdani F, Larroque B, Kapel N, Corcos O, Thomas M, Joly F. Faecal D/L lactate ratio is a metabolic signature of microbiota imbalance in patients with short bowel syndrome.

PLoS ONE 2013; 8(1):e54335. doi:10.1371/journal.pone.0054335. PMID 23372709

18. Vernia P, Caprilli R, Latella G, Barbetti F, Magliocca FM, Cittadini M. Fecal lactate and ulcerative colitis. *Gastroenterology* 1988; 95(6):1564–1568. doi:10.1016/s0016-5085(88)80078-7. PMID 3181680
19. Aguilera L, Campos E, Giménez R, Badía J, Aguilar J, Baldoma L. Dual role of LldR in regulation of the lldPRD operon, involved in L-lactate metabolism in Escherichia coli. *Journal of Bacteriology* 2008; 190(8):2997–3005. doi:10.1128/JB.02013-07. PMID 18263722
20. Yates EA, Philipp B, Buckley C, Atkinson S, Chhabra SR, Sockett RE, Goldner M, Dessaux Y, Cámara M, Smith H, Williams P. N-acylhomoserine lactones undergo lactonolysis in a pH-, temperature-, and acyl chain length-dependent manner during growth of Yersinia pseudotuberculosis and Pseudomonas aeruginosa. *Infection and Immunity* 2002; 70(10):5635–5646. doi:10.1128/IAI.70.10.5635-5646.2002. PMID 12228292
21. Andersen JB, Sternberg C, Poulsen LK, Bjørn SP, Givskov M, Molin S. New unstable variants of green fluorescent protein for studies of transient gene expression in bacteria. *Applied and Environmental Microbiology* 1998; 64(6):2240–2246. doi:10.1128/aem.64.6.2240-2246.1998. PMID 9603842
22. Sobol' IM. Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates. *Mathematics and Computers in Simulation* 2001; 55(1–3):271–280. doi:10.1016/S0378-4754(00)00270-6
23. Saltelli A, Annoni P, Azzini I, Campolongo F, Ratto M, Tarantola S. Variance based sensitivity analysis of model output. Design and estimator for the total sensitivity index. *Computer Physics Communications* 2010; 181(2):259–270. doi:10.1016/j.cpc.2009.09.018
24. Herman J, Usher W. SALib: an open-source Python library for sensitivity analysis. *Journal of Open Source Software* 2017; 2(9):97. doi:10.21105/joss.00097
25. Crank J. The mathematics of diffusion. *Clarendon Press* 1975; 2nd edition, Oxford.
26. Al-Husari M, Webb SD. Theoretical predictions of lactate and hydrogen ion distributions in tumours. *PLoS ONE* 2013; 8(8):e72020. doi:10.1371/journal.pone.0072020
27. Burke JR, Brown P, Quyn A, Lambie H, Tolan D, Sagar P. Tumour growth rate of carcinoma of the colon and rectum: retrospective cohort study. *BJS Open* 2020; 4(6):1200–1207. doi:10.1002/bjs5.50355. PMID 32996713
28. Potten CS, Kellett M, Roberts SA, Rew DA, Wilson GD. Measurement of in vivo proliferation in human colorectal mucosa using bromodeoxyuridine. *Gut* 1992; 33(1):71–78. doi:10.1136/gut.33.1.71. PMID 1740282

29. Wang Q, Wang Y, Ding J, Wang C, Zhou X, Gao W, Huang H, Shao F, Liu Z. A bioorthogonal system reveals antitumour immune function of pyroptosis. *Nature* 2020; 579(7799):421–426. doi:10.1038/s41586-020-2079-1. PMID 32188939
30. Le DT, Uram JN, Wang H, Bartlett BR, Kemberling H, Eyring AD, et al.. PD-1 blockade in tumors with mismatch-repair deficiency. *New England Journal of Medicine* 2015; 372(26):2509–2520. doi:10.1056/NEJMoa1500596. PMID 26028255